

Blood & Immunity (BLIM1041)

Solomon T. Gizaw, PhD
Department of Medical Biochemistry
SoM, CHS, AAU

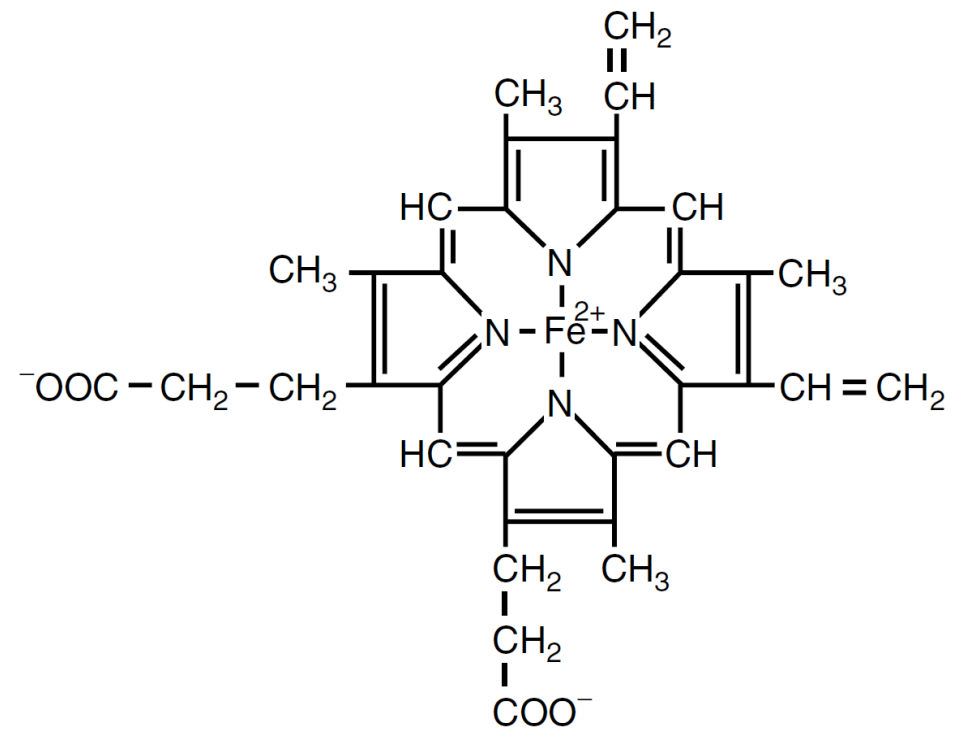
Contents

✓ Heme & Iron Metabolism

✓ Plasma proteins

✓ Blood buffer system

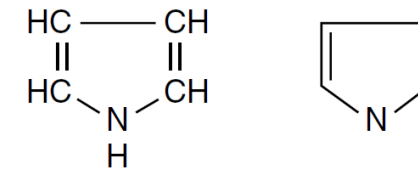
Heme Structure



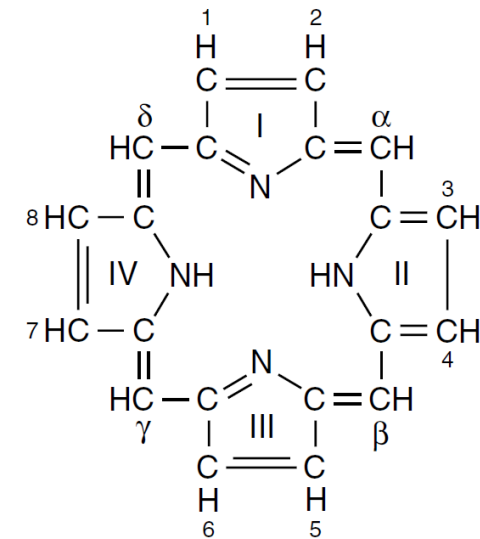
- Heme consists of a porphyrin ring coordinated with an atom of iron.
 - Four pyrrole rings are joined by **methionyl** bridges (—CH—) to form the porphyrin ring.
 - **Eight side chains** serve as substituents on the porphyrin ring, two on each pyrrole. These side chains may be **acetyl** (A), **propionyl** (P), **methyl** (M), or **vinyl** (V) groups.
 - In heme, the order of these groups is **M V M V M P P M**. This order, in which the position of the methyl group is reversed on the fourth ring, is characteristic of the porphyrins of the type III series, the most abundant in nature.
- Heme is the most common porphyrin found in the body.
- It complexes with proteins to form **hemoglobin**, **myoglobin**, and **cytochromes**, including cytochrome P450.

Metalloporphyrins & Hemoporphyrins are important in nature

- Porphyrins are cyclic compounds formed by the linkage of four pyrrole rings through —HC= methenyl bridges.
- A characteristic property of the porphyrins is the formation of complexes with **metal ions** bound to the nitrogen atom of the pyrrole rings.
- ✓ The **iron porphyrins** such as **heme** of hemoglobin
- ✓ The **magnesium**-containing porphyrin **chlorophyll**



Pyrrole



Porphin
(C₂₀H₁₄N₄)

Some important Human and Animal Hemoproteins.

- Proteins that contain heme (hemoproteins) are widely distributed in nature.

Protein	Function
Hemoglobin	Transport of oxygen in blood
Myoglobin	Storage of oxygen in muscle
Cytochrome c	Involvement in electron transport chain
Cytochrome P450	Hydroxylation of xenobiotics
Catalase	Degradation of hydrogen peroxide
Tryptophan pyrrolase	Oxidation of tryptophan

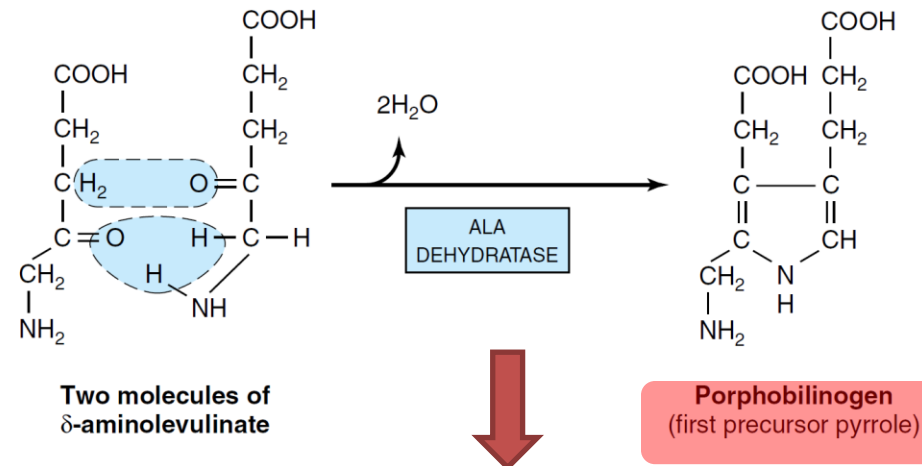
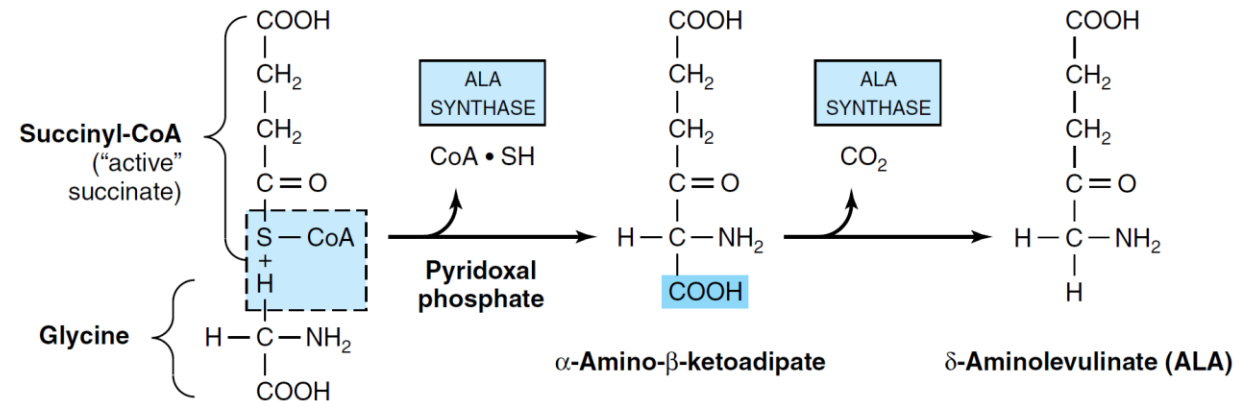
Heme Synthesis

- ✓ ***Heme synthesis** occurs in the precursors of red cells and is a complex pathway that originates from succinyl-coa and glycine.*

*Mutations in any of the steps of heme synthesis lead to a group of diseases known collectively as **porphyrias**.*

Heme is synthesized from Succinyl-CoA & Glycine

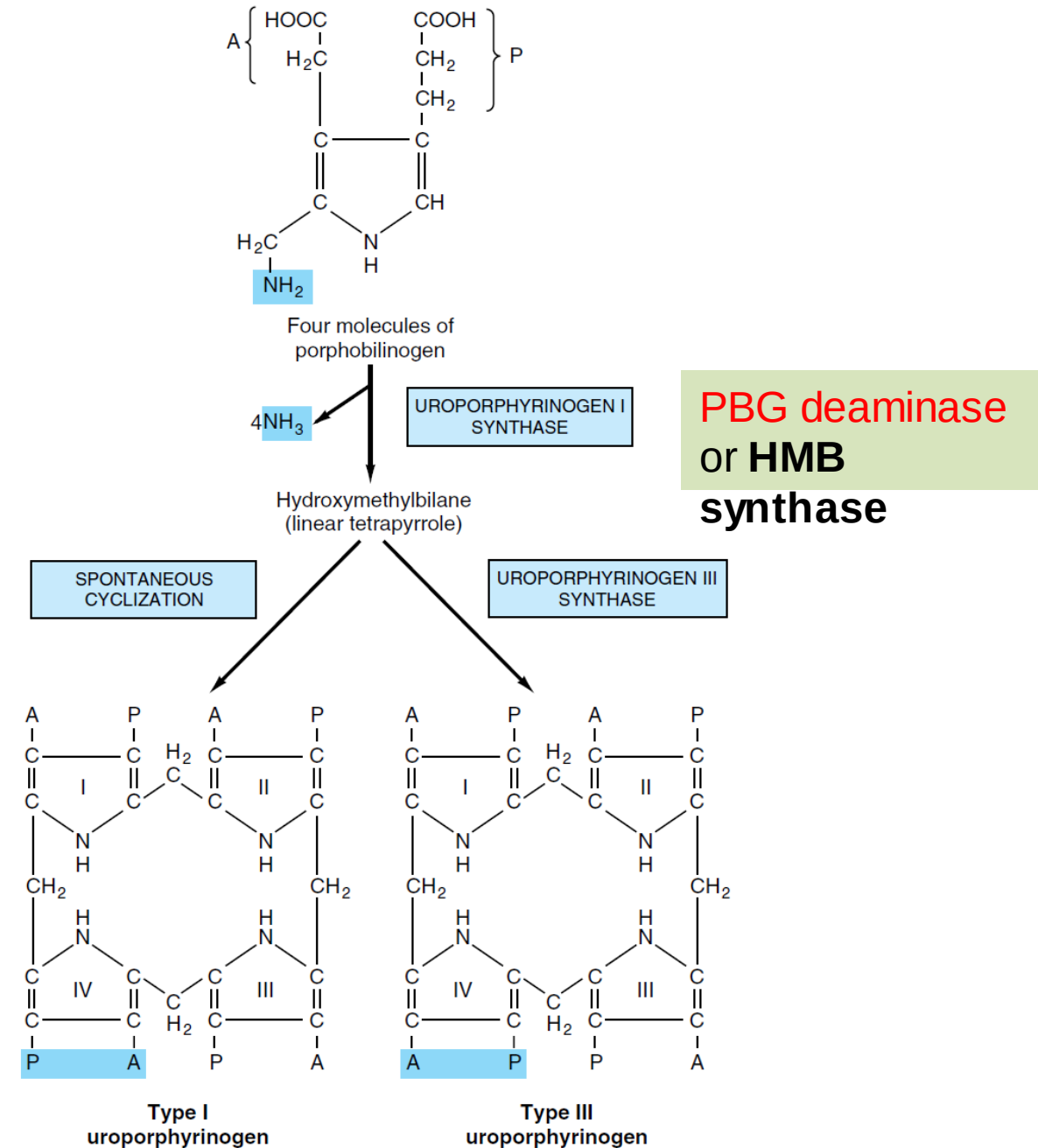
- ✓ Condensation reaction
 - ✓ Mitochondria
 - ✓ **Vitamin B6**
- ✓ Rapid decarboxylation



... is sensitive to inhibition by lead

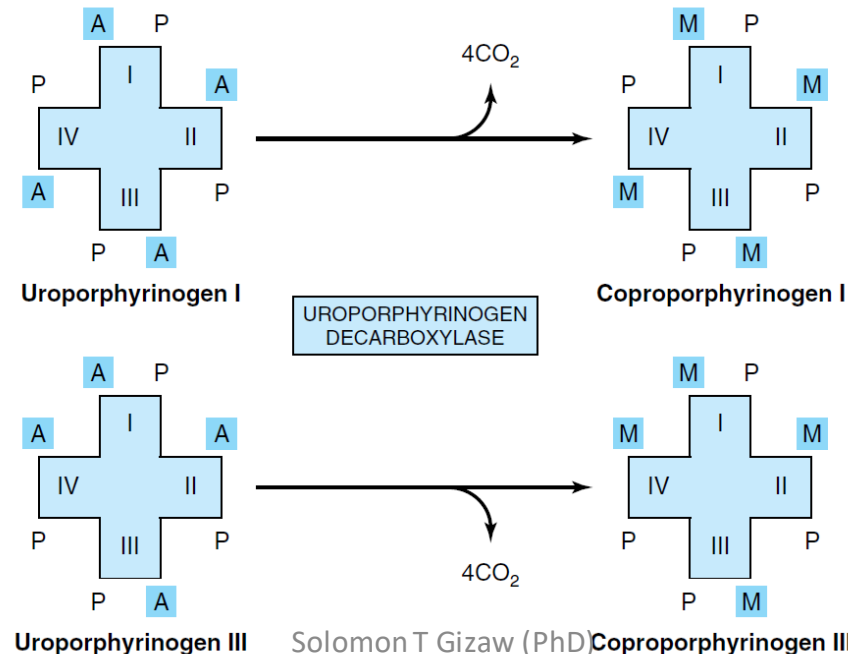
➤ In the **cytosol**, 2 ALA form 2 molecules of water and **Porphobilinogen (PBG)**

- The formation of a cyclic tetrapyrrole—i.e., a **porphyrin**—occurs by condensation of **four** molecules of PBG.
- These four molecules condense in a head-to-tail manner to form a **linear** tetrapyrrole, **hydroxymethylbilane (HMB)**.
- The reaction is catalyzed by **uroporphyrinogen I synthase**, also named **PBG deaminase** or **HMB synthase**.
- Under **normal** conditions, the uroporphyrinogen formed is almost exclusively the **III isomer**, but in certain of the porphyrias, the type I isomers of porphyrinogens are formed in excess.



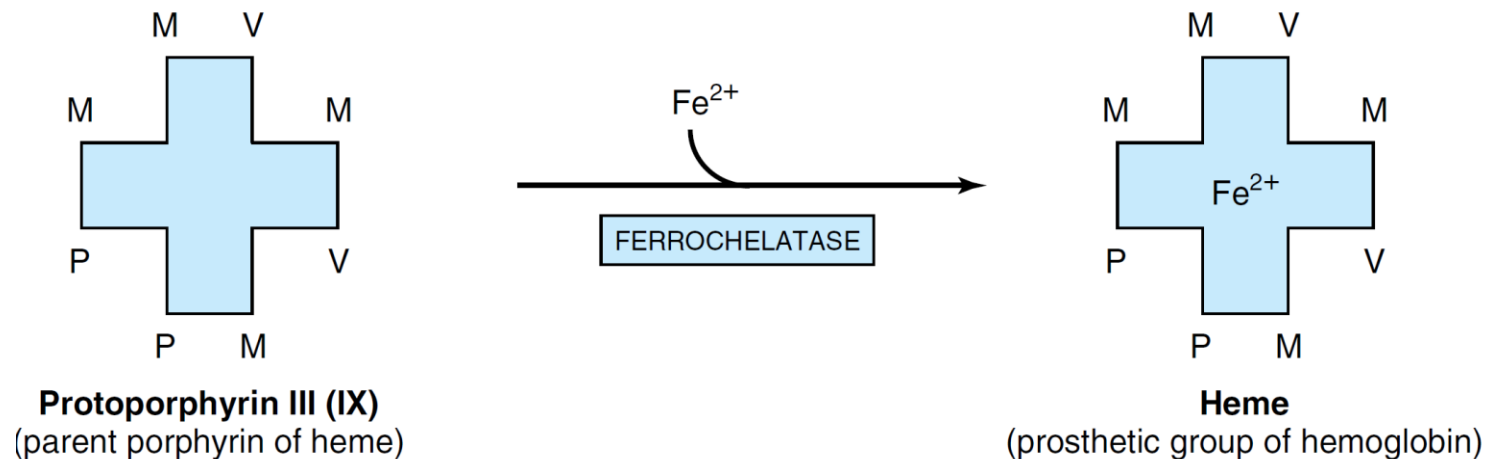
Decarboxylation of uroporphyrinogens to coproporphyrinogens in cytosol

- ✓ Uroporphyrinogen III is converted to coproporphyrinogen III by **decarboxylation** of all of the acetate (**A**) groups, which changes them to methyl (**M**) substituents.
- ✓ The reaction is catalyzed by **uroporphyrinogen decarboxylase**, which is also capable of converting uroporphyrinogen I to coproporphyrinogen I.



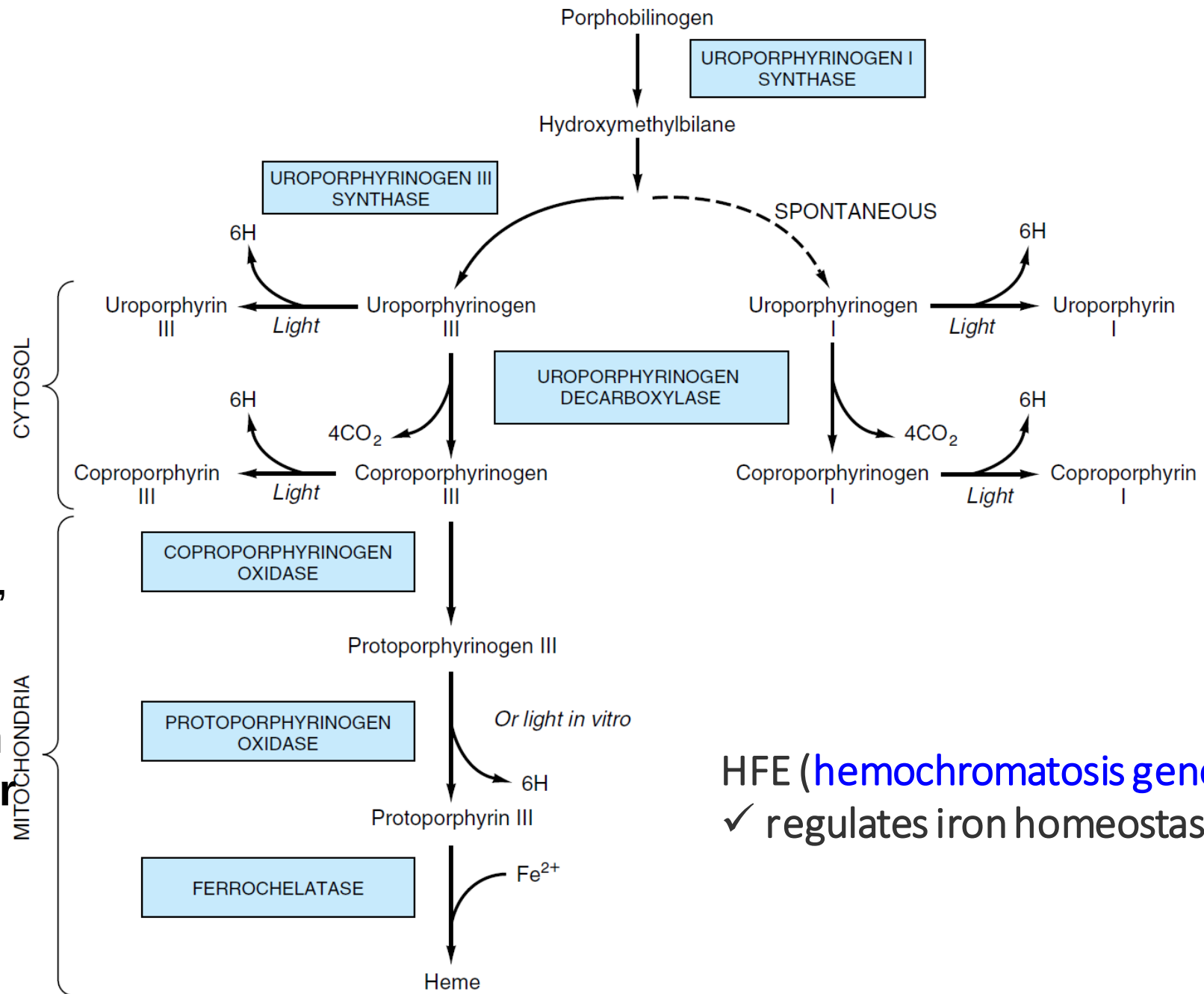
Incorporation of Iron Into Protoporphyrin

- The final step in heme synthesis involves the incorporation of **ferrous iron** into protoporphyrin in a reaction catalyzed by **ferrochelatase (heme synthase)**, another mitochondrial enzyme.



- Fe source
 - ✓ **Meat** (Fe^{2+}): Raw meat to pregnant women (...**Toxoplasmosis**)
 - ✓ Non-heme iron: from plants

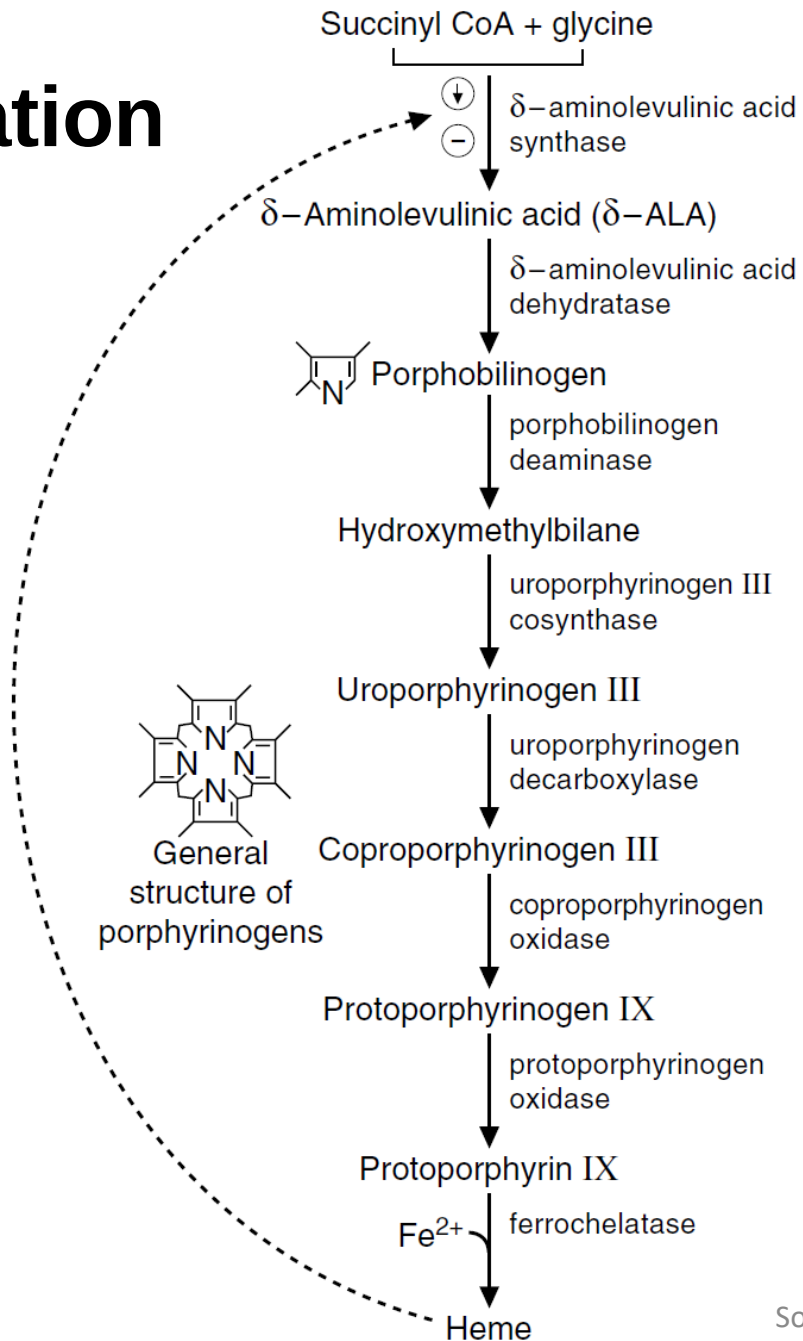
In mammalian liver,
the conversion of
coproporphyrinogen
to protoporphyrin
requires **molecular
oxygen**.



Regulation

- Heme regulates its own synthesis by mechanisms that affect the first enzyme in the pathway, δ -ALA synthase.
- Heme also regulates the synthesis of hemoglobin by stimulating synthesis of the protein globin.
- Heme maintains the ribosomal initiation complex for globin synthesis in an active state.

Regulation

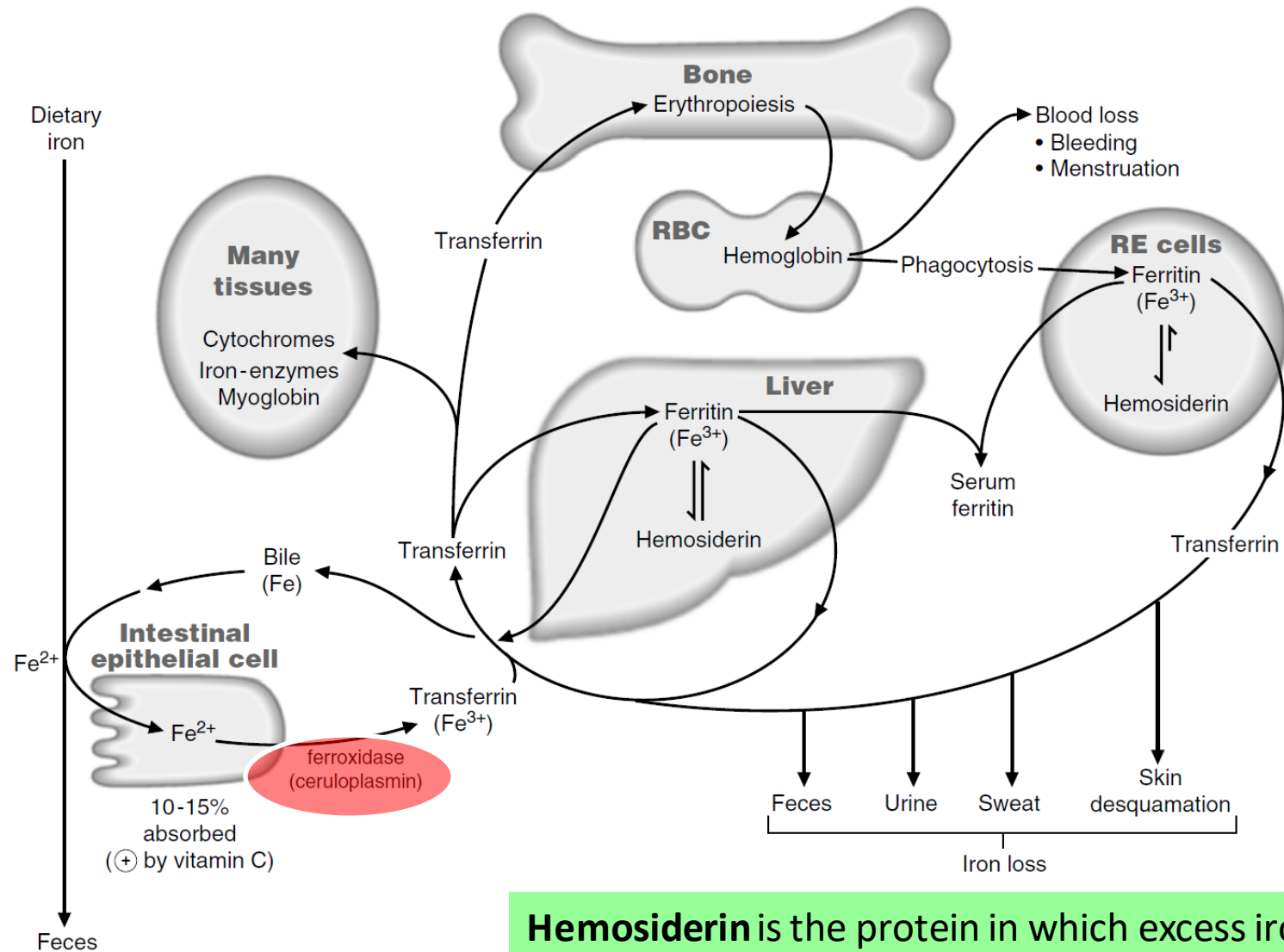


The reaction is catalyzed by **uroporphyrinogen III synthase**,
✓ **PBG deaminase**
✓ **HMB synthase**.

Source of Iron

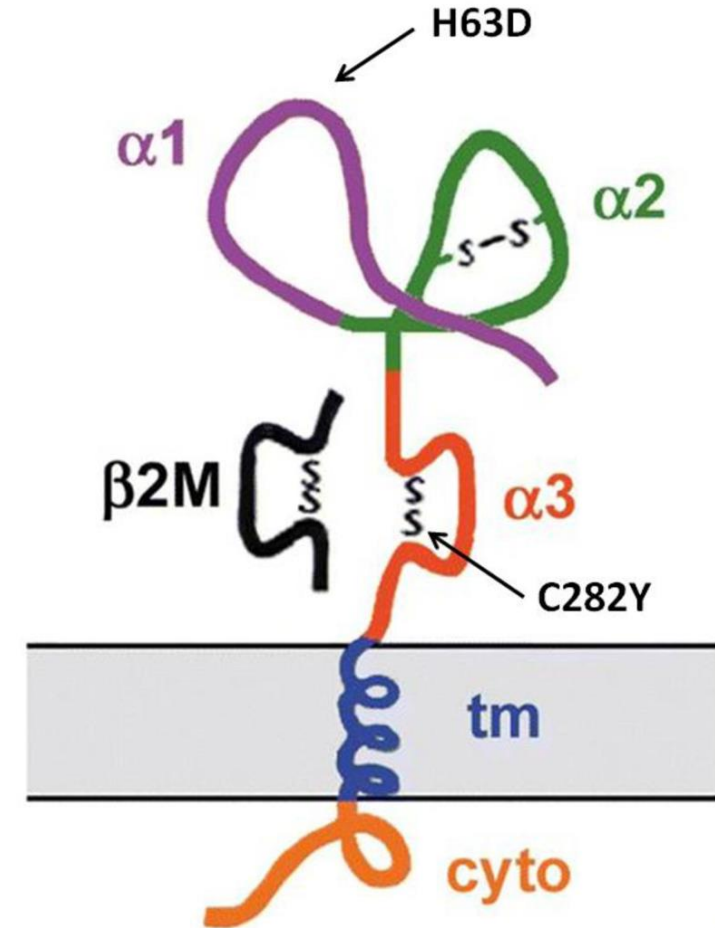
- Iron is absorbed in the ferrous (Fe^{2+}) state, but is oxidized to the ferric state by a **ferroxidase** known as **ceruloplasmin** (a copper-containing enzyme) for transport through the body.
- Because free iron is **toxic**, it is usually found in the body bound to protein.
- Iron is carried in the blood (as Fe^{3+}) by the protein **apotransferrin**, with which it forms a complex known as **transferrin**.
- Transferrin is usually only **one-third saturated** with iron. The total iron-binding capacity of blood, mainly due to its content of transferrin, is approximately 300 g/dL.
- **Storage** of iron occurs in most cells but especially those of the liver, **spleen**, and **bone marrow**.
- In these cells, the storage protein, **apoferritin**, forms a complex with iron (Fe^{3+}) known as **ferritin**.

- ✓ Iron is absorbed from the diet, transported in the blood in **transferrin**, stored in **ferritin**, and used for the synthesis of **cytochromes**, iron-containing enzymes, **hemoglobin**, and myoglobin.



HFE gene (Hemochromatosis gene): homeostatic iron regulator

- HFE protein in association with **beta-2 microglobulin** ($\beta 2M$) at the cell surface.
- The three extracellular domains of HFE are designated $\alpha 1$, $\alpha 2$, and $\alpha 3$.
- $\beta 2M$ is shown associated with the $\alpha 3$ domain.



Hereditary Hemochromatosis (HH)

- The **hemochromatosis gene**, known as HFE, helps regulate the body's absorption of iron.
- Some people can inherit a mutation to this gene that causes their bodies to absorb too much iron.
- Most cases occur due to **C282Y** and **H63D** mutations.
 - ✓ People with **two copies** of C282Y are most likely to have **iron overload**.
 - ✓ People with **one copy of H63D** and **one copy of C282Y** may also have iron overload, but this is **less common** and usually less severe
 - **Iron overload** -- Damage liver (cirrhosis, liver cancer, liver failure), heart, pancreas, endocrine glands, and joints.

Genetic Disorders of the Heme Metabolism: **Porphyrrias**

The Porphyrrias

- The **porphyrias** are a group of disorders due to abnormalities in the pathway of biosynthesis of heme:
 - ✓ genetic
 - ✓ acquired.
- Intermediates of the pathway **accumulate** and may have **toxic effects on the nervous system** that cause neuropsychiatric symptoms.
- When **porphyrinogens accumulate**, they may be converted by light to **porphyrins**, which react with molecular oxygen to form **oxygen radicals**, which may cause severe damage to the skin.
 - Thus, individuals with excessive production of porphyrins are **photosensitive**.

The Porphyrrias

- There's **no cure**, but you can get treatments that will help you manage your symptoms.
- The scarring and increased growth of **facial hair** seen in some porphyrias may have contributed to the development of the **werewolf legends**.
- Types:
 - ✓ Acute porphyria
 - ✓ Cutaneous porphyria

Acute porphyrias

- Primarily affect the **nervous system**.
- Episodes of acute porphyria can cause **abdominal pain, vomiting, constipation, and diarrhea**.
- During an episode, a person may also experience **muscle weakness, seizures, fever, and mental changes** such as anxiety and hallucinations.
 - These signs and symptoms can be **life-threatening**, especially if the muscles that control breathing become paralyzed.
- Acute porphyrias include **acute intermittent porphyria** and **ALAD deficiency porphyria**.

Cutaneous porphyrias

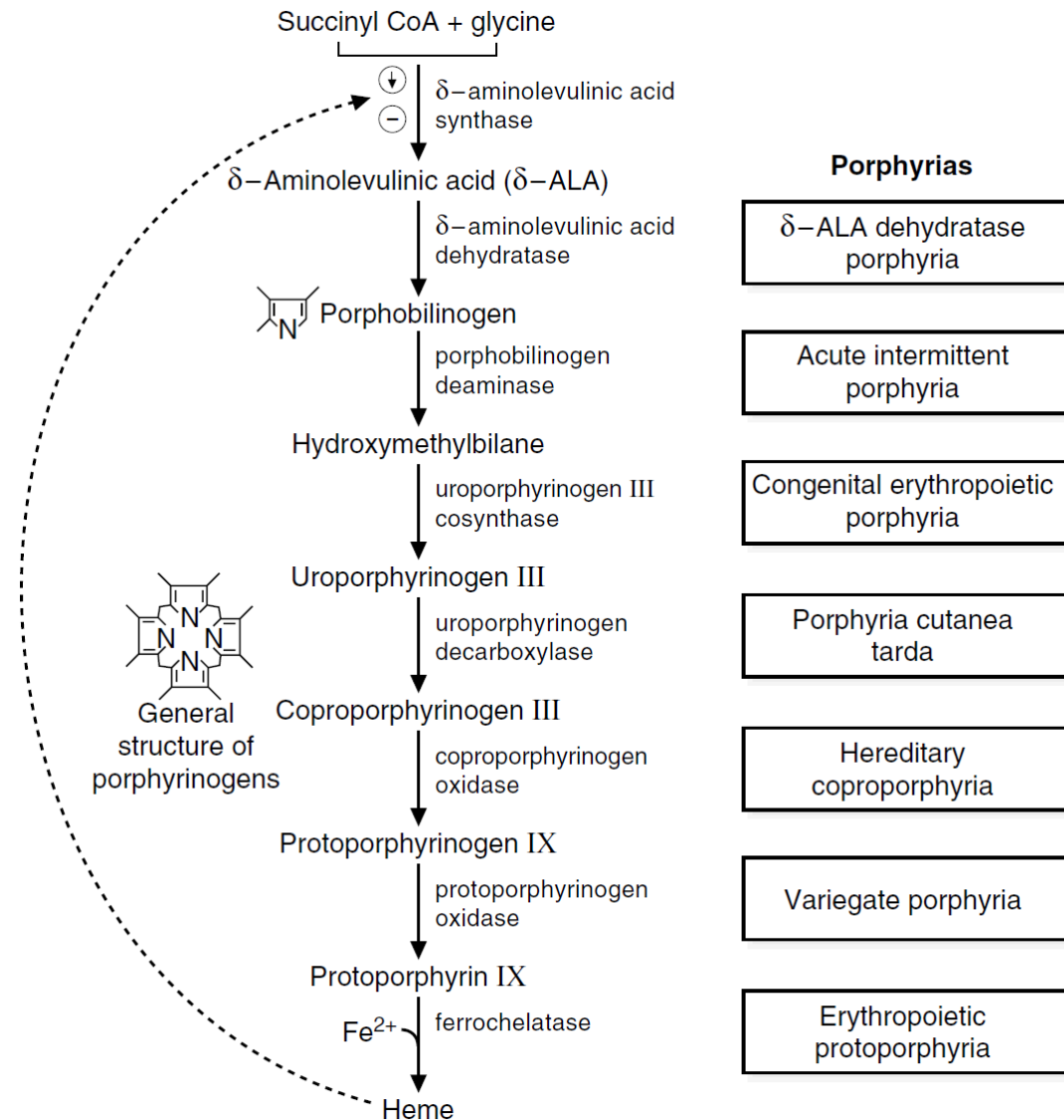
- Primarily affect the **skin**.
- Areas of skin exposed to the sun become fragile and blistered, which can **lead to infection**, scarring, changes in skin coloring (pigmentation), and increased hair growth.
- **Cutaneous porphyrias include**
 - ✓ Congenital erythropoietic porphyria (**Uroporphyrinogen III synthase**)
 - ✓ Hepatoerythropoietic porphyria
 - ✓ Horphyria cutanea tarda (**Uroporphyrinogen decarboxylase**)
 - ✓ Erythropoietic protoporphyria (**ferrochelatase**)

Causes

- **Mutations** in most of these genes reduce enzyme activity, which limits the amount of heme the body can produce.
- **As a result, compounds called porphyrins and porphyrin precursors, which are formed during the process of heme production, can build up abnormally in the liver and other organs.**
- When these substances accumulate in the skin and interact with sunlight, they cause **cutaneous forms of porphyria.**
- The acute forms of the disease occur when porphyrins and porphyrin precursors build up in and **damage the nervous system.**

Causes

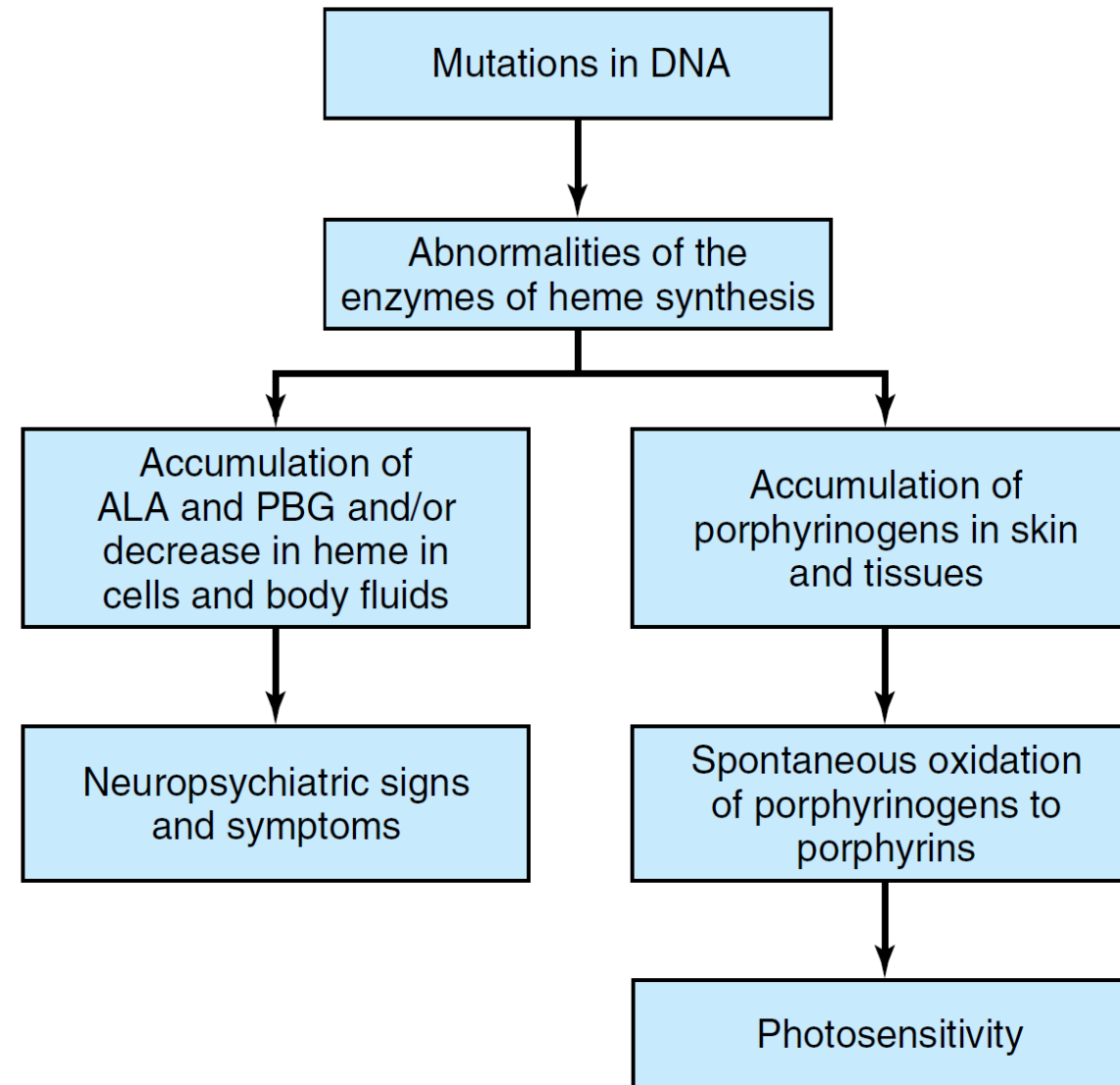
Gene	Protein
ALAS2	5'-aminolevulinate synthase 2
ALAD	aminolevulinate dehydratase
HMBS	Hydroxymethylbilane synthase
UROS	uroporphyrinogen III synthase
UROD	uroporphyrinogen decarboxylase
CPOX	coproporphyrinogen oxidase
PPOX	protoporphyrinogen oxidase
FECH	ferrochelatase



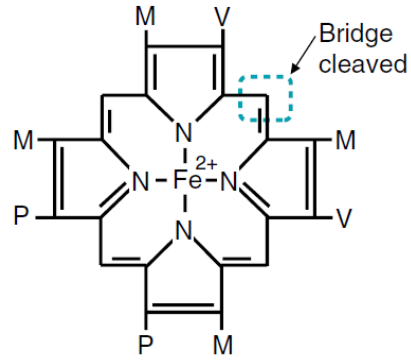
Summary of major findings in the porphyrias

Enzyme Involved ²	Type, Class, and MIM Number	Major Signs and Symptoms	Results of Laboratory Tests
1. ALA synthase (erythroid form)	X-linked sideroblastic anemia ³ (erythropoietic) (MIM 201300)	Anemia	Red cell counts and hemoglobin decreased
2. ALA dehydratase	ALA dehydratase deficiency (hepatic) (MIM 125270)	Abdominal pain, neuropsychiatric symptoms	Urinary δ -aminolevulinic acid
3. Uroporphyrinogen I synthase ⁴	Acute intermittent porphyria (hepatic) (MIM 176000)	Abdominal pain, neuropsychiatric symptoms	Urinary porphobilinogen positive, uroporphyrin positive
4. Uroporphyrinogen III synthase	Congenital erythropoietic (erythropoietic) (MIM 263700)	No photosensitivity	Uroporphyrin positive, porphobilinogen negative
5. Uroporphyrinogen decarboxylase	Porphyria cutanea tarda (hepatic) (MIM 176100)	Photosensitivity	Uroporphyrin positive, porphobilinogen negative
6. Coproporphyrinogen oxidase	Hereditary coproporphyria (hepatic) (MIM 121300)	Photosensitivity, abdominal pain, neuropsychiatric symptoms	Urinary porphobilinogen positive, urinary uroporphyrin positive, fecal protoporphyrin positive
7. Protoporphyrinogen oxidase	Variegate porphyria (hepatic) (MIM 176200)	Photosensitivity, abdominal pain, neuropsychiatric symptoms	Urinary porphobilinogen positive, fecal protoporphyrin positive
8. Ferrochelatase	Protoporphyria (erythropoietic) (MIM 177000)	Photosensitivity	Fecal protoporphyrin positive, red cell protoporphyrin positive

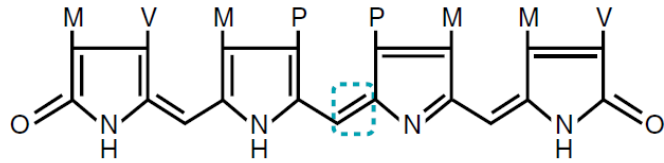
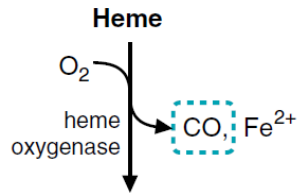
Biochemical causes of the major signs and symptoms of porphyrias



Degradation of Heme

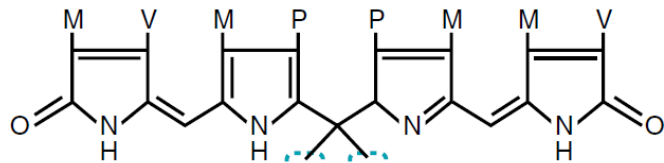
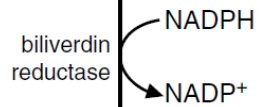


Heme oxygenase

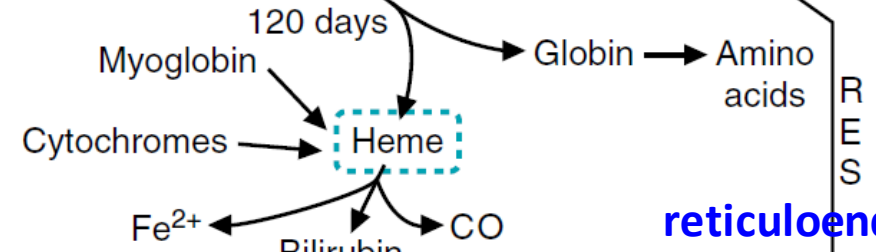
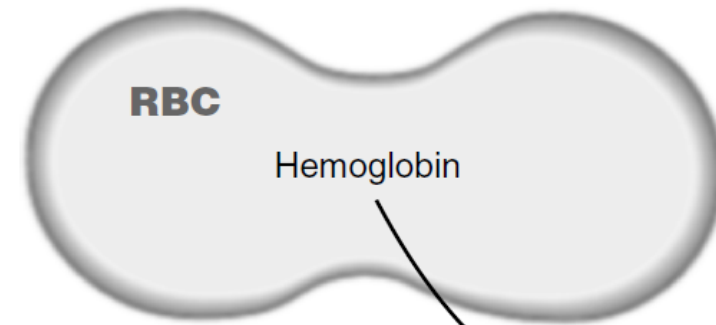


Biliverdin IX α

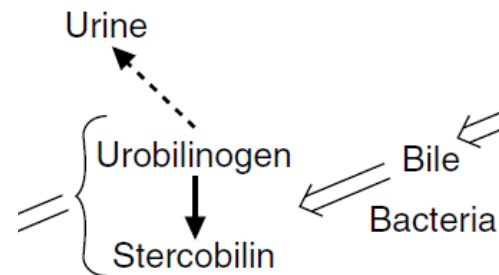
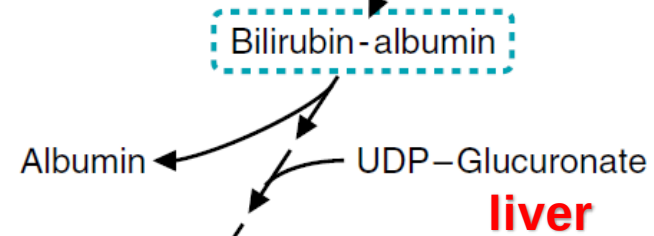
Biliverdin reductase



Bilirubin IX α



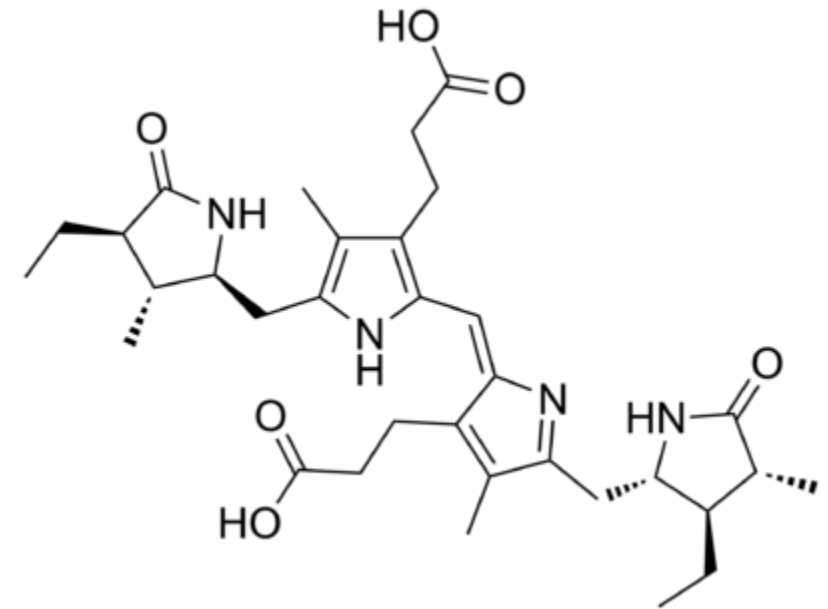
reticuloendothelial system



➤ This conjugated form of bilirubin is excreted into the bile.

Degradation of Heme

- In the intestine, bacteria **deconjugate** bilirubin diglucuronide and convert the bilirubin to **urobilinogens**.
- Some urobilinogen is absorbed into the blood and excreted in the urine.
- However, most of the urobilinogen is oxidized to **urobilins**, such as **stercobilin**, and excreted in the feces.
 - ✓ These pigments give feces the **brown color** of human feces.
 - ✓ Stercobilin and urobilin can be used as a marker for biochemical identification of fecal pollution levels in rivers.

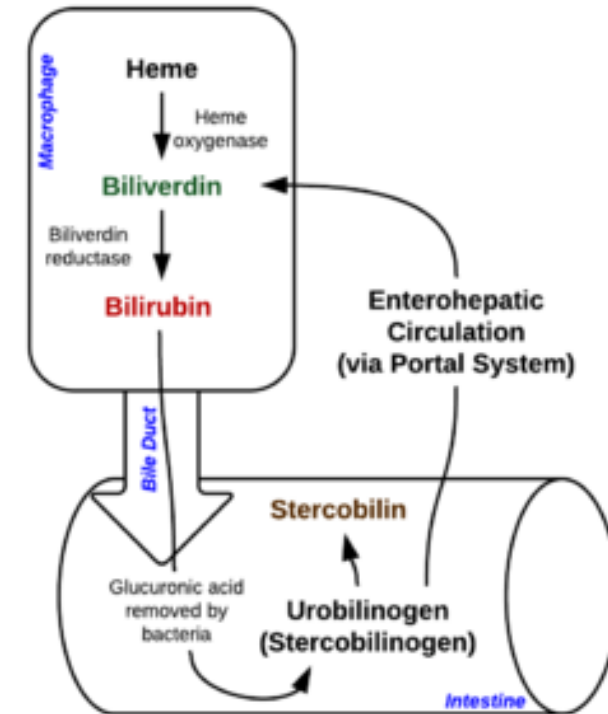


stercobilin

Catabolism of Heme Produces Bilirubin

- Bilirubin

- ✓ 0-0.3 mg/dl - Direct (conjugated) bilirubin level
- ✓ 0.1-1.2 mg/dl - Total serum bilirubin level



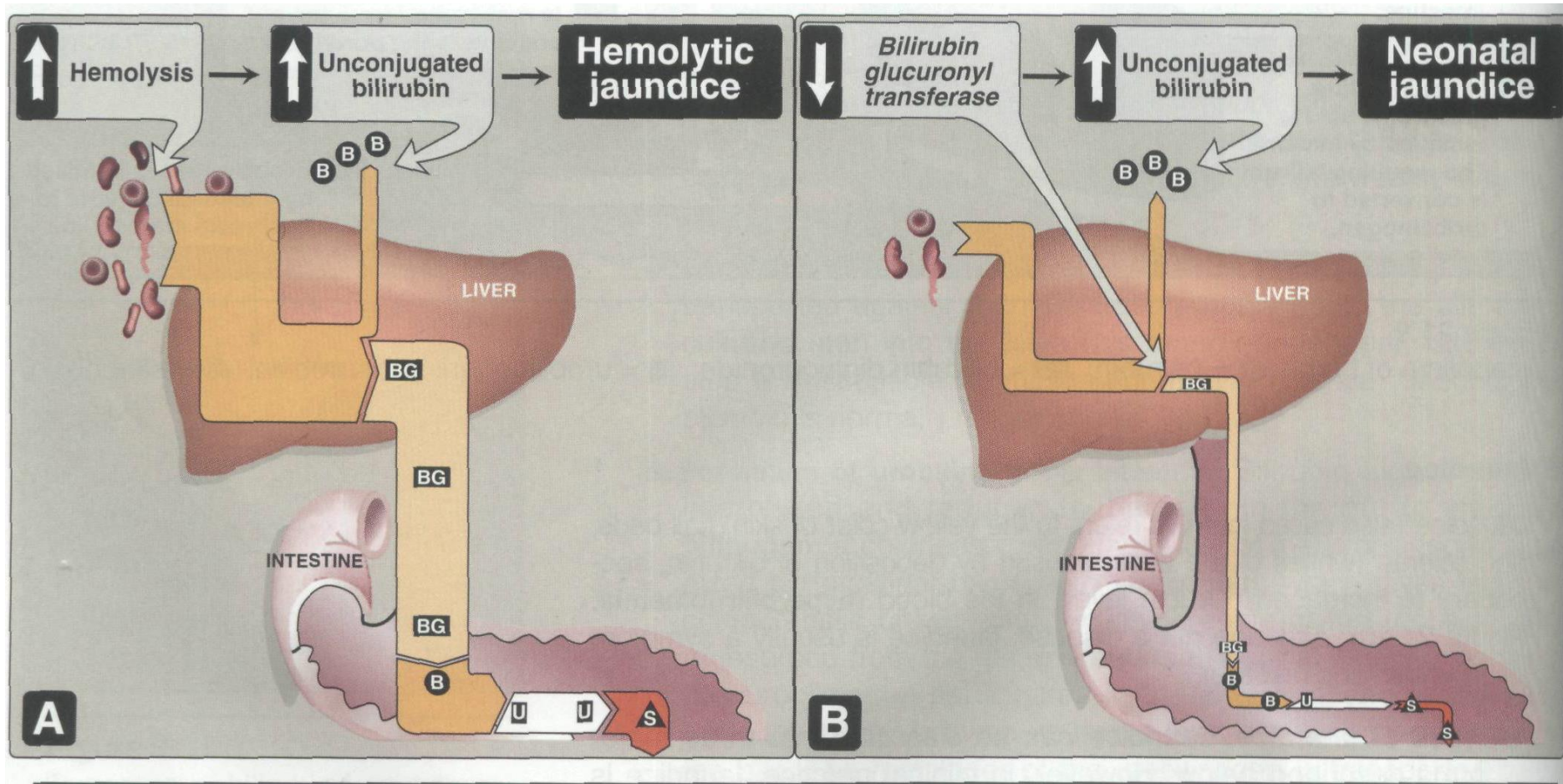
- Hyperbilirubinemia & -Jaundice (2–3 mg/dL (34-51 $\mu\text{mol/L}$) typically results in jaundice)

Jaundice(icterus)

- The yellow color of skin, nail beds, and sclerae (whites of the eyes) caused by depositon of bilirubin, secondary to increased bilirubin levels in the blood (hyperbilirubineia).
- Although not a diseases, jaundice is usually a symptom of an underlying disorder.
- Types:
 - ✓ In clinical practice, it is often more complex than indicated in the this simple classification; however, it classified into three major forms
- **Types**
 - ✓ **Hemolytic jaundice**
 - ✓ **Obstructive jaundice**
 - ✓ **Hepatocellular jaundice**

- **Hemolytic jaundice:** The liver has the capacity to conjugate and excrete over 3000 mg of bilirubin per day, whereas the normal production of bilirubin is only 300 mg/day.
- This excess allows the liver to respond to increased heme degradation **with** a corresponding increase in conjugation and secretion of **bilirubin diglucuronide**.
- However, massive lysis of red blood cells (for example, in patients with sickle cell anemia, pyruvate kinase or glucose 6-phosphate dehydrogenase deficiency, malaria) may produce bilirubin faster than it can be conjugated.
- More bilirubin is excreted into the bile, the amount of bilirubin entering the enterohepatic circulation is increased, and urinary urobilinogen is increased. Unconjugated bilirubin levels become elevated in the blood, causing jaundice.

Alteration in the metabolism of heme



BG: Bilirubin glucouronide; B: Bilirubin; U: Urobilinogen; and S: Stercobilin

- **Obstructive jaundice:** In this instance, jaundice is not caused by overproduction of bilirubin, but instead results from obstruction of the bile duct.
- For example, the presence of a hepatic tumor or bile stones may block the bile ducts, preventing passage of bilirubin into the intestine.
- Patients with obstructive jaundice experience gastrointestinal pain and nausea, and produce stools that are a pale, clay color.
- The liver "regurgitates" conjugated bilirubin into the blood (hyperbilirubinemia). The compound is eventually excreted in the urine.
- Prolonged obstruction of the bile duct can lead to liver damage and a subsequent rise in unconjugated bilirubin.]

- **Hepatocellular jaundice:** Damage to liver cells (for example, in patients with cirrhosis or hepatitis) can cause unconjugated bilirubin levels to increase in the blood as a result of decreased conjugation.
- The bilirubin that is conjugated is not efficiently secreted into the bile but instead diffuses ("leaks") into the blood.
- Urobilinogen is increased in the urine because hepatic damage decreases the enterohepatic circulation of this compound, allowing more to enter the blood, from which it is filtered into the urine.
- The urine thus becomes dark in color, whereas stools are a pale, clay color.
- Plasma levels of AST(SGOT) and ALT are elevated, and the patient experiences nausea and anorexia.

- **Jaundice in newborns:** Newborn infants, particularly premature babies, often accumulate bilirubin, because the activity of hepatic *bilirubin glucuronyl transferase* is low at birth – reaches adult levels in about four weeks (Figures 21.11B and 21.12).
- Elevated bilirubin, in excess of the binding capacity of albumin, can diffuse into basal ganglia and cause toxic encephalopathy (**kernicterus**).
- Thus, newborns with significantly elevated bilirubin levels are treated with blue fluorescent light, which converts bilirubin to more polar and, hence, water-soluble isomers. These photoisomers can be excreted into the bile without conjugation to glucuronic acid.

Contents

✓ Heme & Iron Metabolism

✓ Plasma proteins

Blood Plasma Proteins

- *The blood is the body's main transport system.*
- *Although the transport and delivery of oxygen to the cells of the tissues is carried out by specialized cells, **other vital components** such as **nutrients, metabolites, electrolytes, and hormones**, are all **carried in the noncellular fraction of the blood**, the **plasma**.*
- *Some components, such as glucose, are dissolved in the plasma; others, for example, lipids and steroid hormones, are bound to carrier proteins for transport.*
- *The **osmotic pressure** of the plasma proteins regulates the **distribution of water** between the blood and the tissues.*

I

Plasma proteins maintain proper distribution of water between blood & tissues

- ✓ Body Fluid Maintenance between Tissues and Blood
- ✓ The Major Serum Protein, Albumin

✓ Body Fluid Maintenance between Tissues and Blood

- When the **cells** are removed from the blood, the remaining plasma is composed of **water**, **nutrients**, **metabolites**, hormones, electrolytes, and **proteins**.
- Plasma has essentially the same electrolyte composition as other extracellular fluids and constitutes approximately **25%** of the body's total extracellular fluid.
- Plasma proteins in conjunction with platelets maintain the **integrity of the circulatory system** through the process of clotting.
- The plasma proteins serve a number of functions:
 - ✓ *maintaining the proper distribution of water between the blood and the tissues,*
 - ✓ *transporting nutrients, metabolites, and hormones throughout the body,*
 - ✓ *defending against infection,*
 - ✓ *maintaining the integrity of the circulation through clotting.*

✓ Body Fluid Maintenance between Tissues and Blood

- Many diseases alter the amounts of plasma proteins produced and, hence, their concentration in the blood.
- These changes can be determined by **electrophoresis** of plasma proteins over the course of a disease.

The Major Serum Protein, Albumin

- The liver produces a number of transport or binding proteins and secretes them into the blood.
- The major protein synthesized is albumin, which constitutes approximately 60% of the total plasma protein, but because of its relatively small size (69 kDa) is thought to contribute 70 to 80% of the total osmotic pressure of the plasma.
- Albumin, like most plasma proteins, is a glycoprotein and is a carrier of free fatty acids, calcium, zinc, steroid hormones, copper, and bilirubin.

Specific Plasma Binding Proteins Synthesized in the Liver

Ceruloplasmin	Binds copper; appears to be more important as a copper storage pool than as a transport protein; integrates iron and copper homeostasis
Corticosteroid-binding globulin	Binds cortisol
Haptoglobin	Binds extracorporeal heme
Lipoproteins	Transport cholesterol and fatty acids
Retinol-binding protein	Binds vitamin A
Sex hormone-binding globulin	Binds estradiol and testosterone
Transferrin	Transports iron
Transthyretin	Binds thyroxine (T ₄); also forms a complex with retinol-binding protein

Blood Plasma Proteins

- The major protein synthesized is albumin, which constitutes approximately **60% of the total plasma protein**, but because of its relatively **small size** (69 kDa) is thought to contribute **70 to 80%** of the total osmotic pressure of the plasma.
- **Albumin**, like most plasma proteins, is a glycoprotein and is a carrier of free fatty acids, calcium, zinc, steroid hormones, copper, and bilirubin.
- Many drugs also bind to albumin, which may have important pharmacologic implications.
- For example, when a drug binds to albumin, such binding will likely **lower the effective concentration of that drug** and may lengthen its lifetime in circulation.
- Drug dosimetry may need to be recalculated if a patient's plasma protein concentration is abnormal.

II

The plasma contains
proteins that aid in
Immune defense

- ✓ Two different sets of proteins aid in the immune response, the immunoglobulins and the complement proteins.

Bone marrow

- **Hematopoietic stem cells:** Myeloid and lymphoid stem cells are two types of multipotent stem cells in the bone marrow that give rise to different blood cells.
 - ✓ Myeloid stem cells
 - Produce **red blood cells**, **platelets**, and some types of **white blood cells**, such as monocytes, macrophages, neutrophils, basophils, and eosinophils.
 - ✓ Lymphoid stem cells.
 - Produce the types of white blood cells classified as **lymphocytes**, such as **T cells**, **B cells**, **natural killer cells**, and **innate lymphoid cells**.
- Myeloid cells can mature in the bone marrow, while lymphoid cells require maturation outside the bone marrow.

- **Lymphoid stem cells** can become three types of white blood cells that play critical roles in your adaptive immunity:

1) Natural killer cells: contain **enzymes** to destroy cancer cells or viruses

2) T cells: destroy infected cells and activate other immune cells

3) B cells: become **plasma cells** when activated by foreign invaders and produce Immunoglobulins.

Plasma cells

- Plasma cells are activated B cells.
- They have the capacity to generate mass quantities of **immunoglobulins** that are programmed to recognize a specific epitope (foreign molecule).

The plasma contains proteins that aid in Immune defense

- The immunoglobulins are secreted by a subset of **differentiated B lymphocytes** termed plasma cells and bind antigens at binding sites formed by **the hypervariable regions** of the proteins.
- Once the antibody–antigen complex is formed, it must be **cleared from circulation**.
 - ✓ The complement system participates in this function.

Immunoglobulin

- ✓ An antibody (Ab), also known as an **immunoglobulin (Ig)**, is a **large**, Y-shaped protein used by the immune system to identify and neutralize foreign objects such as **pathogenic bacteria** and **viruses**.
- ✓ Made up of sugar and proteins produced by plasma cells, a type of WBC that help your body defend against infections and cancer.
- ✓ This binding helps neutralize the foreign cell and signals to your white blood cells to **destroy** them.

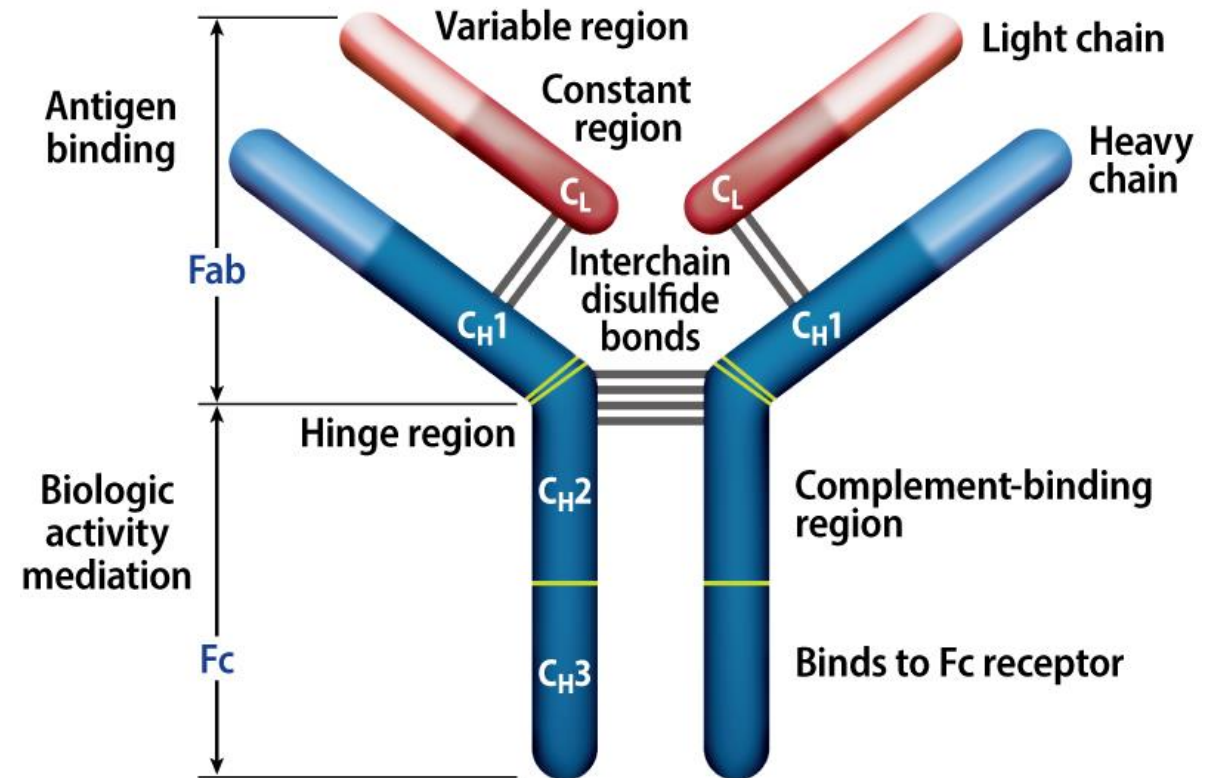
✓

- **Three main functions:**

- 1) preventing foreign invaders from entering cells by neutralizing them.
- 2) coating foreign invaders to tag them for destruction by white blood cells called macrophages.
- 3) triggering the destruction of foreign invaders by **stimulating other immune responses**.

Immunoglobulin (Ig)

- ✓
- ✓ Two light chains: kappa (κ) or lambda (λ).
- ✓ Two heavy chains: IgG, IgA, IgD, IgE, and IgM.



Types of Immunoglobulin (Ig) : classification based on heavy chains

- **IgM:** Provides **rapid defense** against infectious diseases.
- **IgG:** Provides most of the immunoglobulin-based immunity against foreign molecules but has a slower effect than IgM.
- **IgA:** Provides defense against infectious agents and is found mostly in your **mucus membranes**.
 - ✓ It protects the surfaces of your **respiratory, digestive, and urinary systems**.
- **IgE:** Provides protection against **parasites**.
 - ✓ It also **triggers histamine release** from WBCs called **mast cells** and basophils to trigger allergic reactions.
- **IgD:** Has a largely unknown function.
 - ✓ It may serve as a **receptor on B cells** that haven't been previously exposed to foreign molecules.

Immunoglobulins in medicine

- Immunoglobulin levels are useful biomarkers for monitoring many different diseases, including
 - ✓ immunodeficiencies
 - ✓ autoimmune diseases

Low and High immunoglobulin levels

- Low immunoglobulin levels suggest that your immune system **is not functioning optimally**.
- Conditions associated with low immunoglobulin levels include:
 - ✓ serious infections
 - ✓ medication side effects
 - ✓ kidney disease
 - ✓ burns
 - ✓ Malnutrition
- Elevated immunoglobulin levels are seen in people with conditions such as:
 - ✓ autoimmune disease
 - ✓ hepatitis
 - ✓ cirrhosis
 - ✓ chronic infection
 - ✓ some cancers like multiple myeloma
 - ✓ allergies
 - ✓ inflammatory bowel disease

Antibody deficiency disorders

- Some people **do not produce immunoglobulins** or **do not produce as many immunoglobulins** as they typically should.
- They may present shortly after birth with **recurrent infections**.
- These conditions are often treated with **intravenous immune globulin (IVIG)** where immunoglobulins are infused through a vein.
- **IVIG is often used to treat these conditions**

Condition	Description
immune thrombocytopenia	autoimmune destruction of platelets
autoimmune hemolytic anemia	autoimmune destruction of red blood cells
Guillain-Barré syndrome	autoimmune destruction of nerves that control breathing
chronic inflammatory demyelinating polyneuropathy	autoimmune destruction of your nerves' protective coating

Immunoglobulin blood tests

Immunoglobulin test	Biological range	Low when?	High when?
Ig A	0.8 to 3.0g/L	?	?
IgD	0.003 to 0.03g/L	?	?
IgE	0.0002 to 0.002g/L	?	?
IgG	6.0 to 16.0g/L	?	?
IgM	0.4 to 2.5g/L	?	?

II

The plasma contains proteins that aid in Immune defense

- ✓ Two different sets of proteins aid in the immune response, the **immunoglobulins** and the complement proteins.

Plasma Protein Mediators

- Many nonspecific innate immune factors are found in plasma, the fluid portion of blood.
- Plasma contains electrolytes, sugars, lipids, and proteins, each of which helps to maintain homeostasis (i.e., stable internal body functioning), and contains **the proteins involved in the clotting of blood**.
- Additional proteins found in blood plasma, such as
 - ✓ **acute-phase proteins**,
 - ✓ **complement proteins**, and
 - ✓ **cytokines**, are involved in the nonspecific innate immune response.

•

Acute-Phase Proteins

- The acute-phase proteins are another class of **antimicrobial mediators**.
- Acute-phase proteins are primarily produced in the **liver** and secreted into the blood **in response to inflammatory molecules** from the immune system.

C-reactive protein	Coats bacteria (opsonization), preparing them for ingestion by phagocytes
Serum amyloid A	
Ferritin	Bind and sequester iron, thereby inhibiting the growth of pathogens
Transferrin	
Fibrinogen	Involved in formation of blood clots that trap bacterial pathogens
Mannose-binding lectin	Activates complement cascade

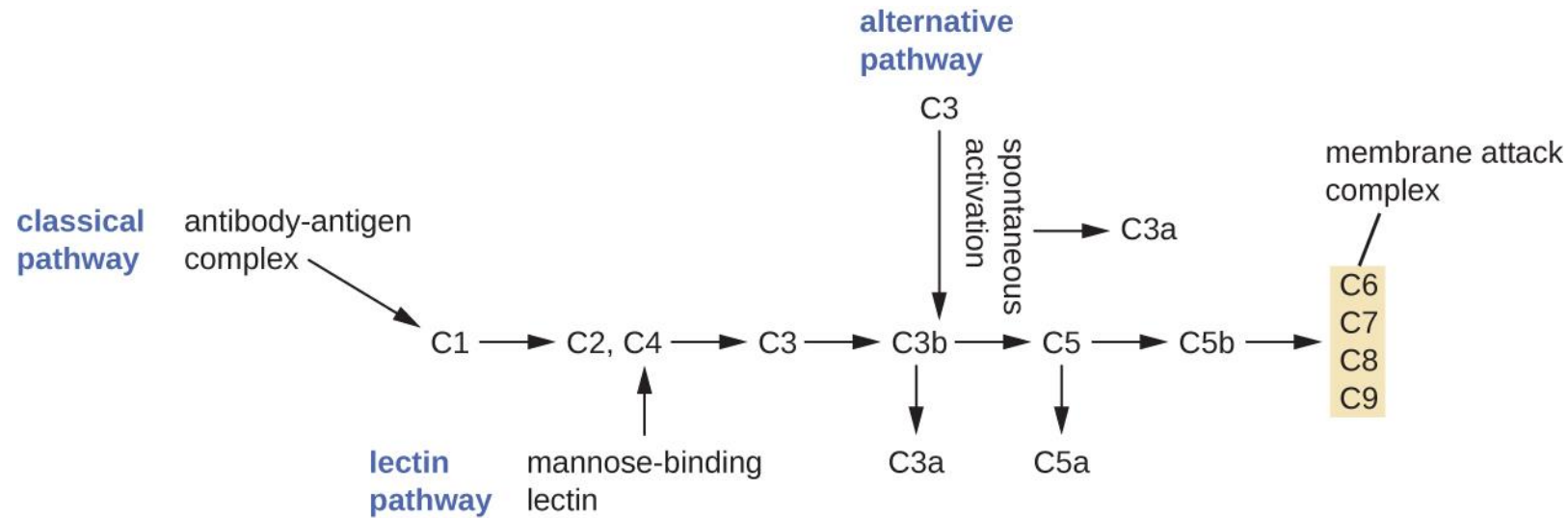
The Complement System

- ✓ The complement system is a group of plasma protein mediators that can act as an innate nonspecific defense while also serving to connect innate and adaptive immunity.
- These precursor proteins become activated when stimulated or triggered by a variety of factors, including the presence of microorganisms.
- ✓ The complement system is composed of more than 30 proteins (including C1 through C9) that normally circulate as precursor proteins in blood.

The plasma contains proteins that aid in Immune defense

- Activation of the complement system by either trigger results in a **proteolytic activation cascade of the proteins of the complement system**, resulting in the release of biologically **active peptides** or polypeptide fragments.
- These **peptides**
 - ✓ mediate the inflammatory response,
 - ✓ attract phagocytic cells to the area,
 - ✓ initiate degranulation of granulocytes, and
 - ✓ promote clearance of antigen–antibody complexes.

- ✓ The process by which circulating complement precursors become functional is called **complement activation**.
- ✓ This process is a cascade that can be triggered by one of three different mechanisms, known as the **alternative**, **classical**, and **lectin pathways**.
- ✓ all three result in the activation of the complement protein C3, which produces C3a and C3b.



- ✓ C5b also binds to the cell surface and then recruits C6 through C9; these molecules form a ring structure called the **membrane attack complex (MAC)**, which punches through the cell membrane of the invading pathogen, causing it to **swell and burst**.

The Complement System

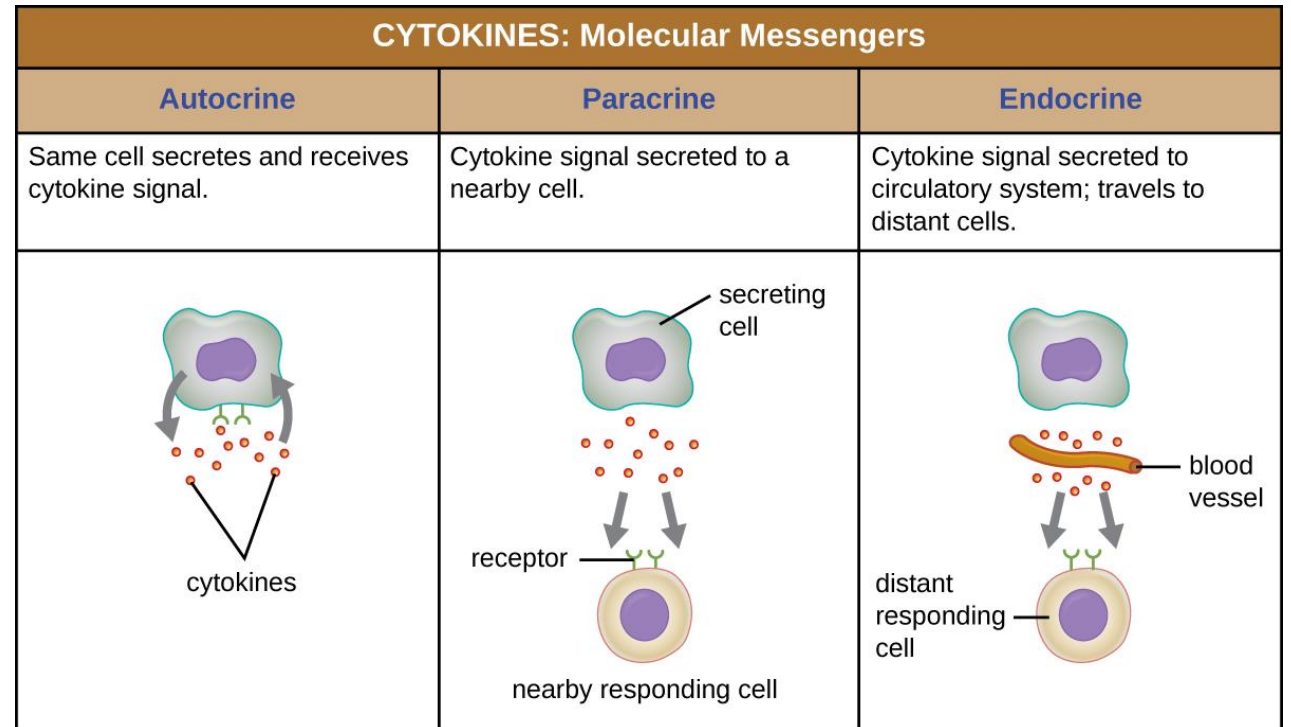
- Although each complement activation pathway is initiated in a different way, they all provide the same **protective outcomes**:
 - ✓ opsonization,
 - ✓ inflammation,
 - ✓ chemotaxis, and
 - ✓ cytolysis.
- Opsonization refers to **the coating of a pathogen by a chemical substance** (called an opsonin) that allows phagocytic cells to **recognize, engulf, and destroy** it more easily.
 - ✓ Opsonins from the complement cascade include C1q, C3b, and C4b.
 - ✓ Additional important opsonins include **mannose-binding proteins** and **antibodies**.
- The complement fragments C3a and C5a are well-characterized anaphylatoxins (**complement peptides**) with **potent proinflammatory functions**.
- Anaphylatoxins activate mast cells, causing **degranulation (release of substances)** and the release of inflammatory chemical signals, including mediators that cause **vasodilation** and **increased vascular permeability**. 0978071641

The Complement System

- **protective outcomes:**
 - ✓ chemotaxis,
 - is the movement of an organism or entity in response to a chemical stimulus.
 - Migration of **leukocytes** during injury or infection.
- Cytokines: induce directional movement of leukocytes.
- C5a is also one of the most potent **chemoattractants** for neutrophils and other white blood cells, cellular defenses.
- Occurs when a **cell bursts** and releases its contents into the extracellular environment due to a great influx of water into the cell, far exceeding the capacity of the cell membrane to contain the extra volume.

Cytokines

- are soluble proteins that act as communication signals between cells.
- In a nonspecific innate immune response, various cytokines may be released to stimulate the production of chemical mediators or other cell functions, such as cell proliferation, cell differentiation, inhibition of cell division, apoptosis, and chemotaxis.



Cytokines

Three important classes of cytokines are the

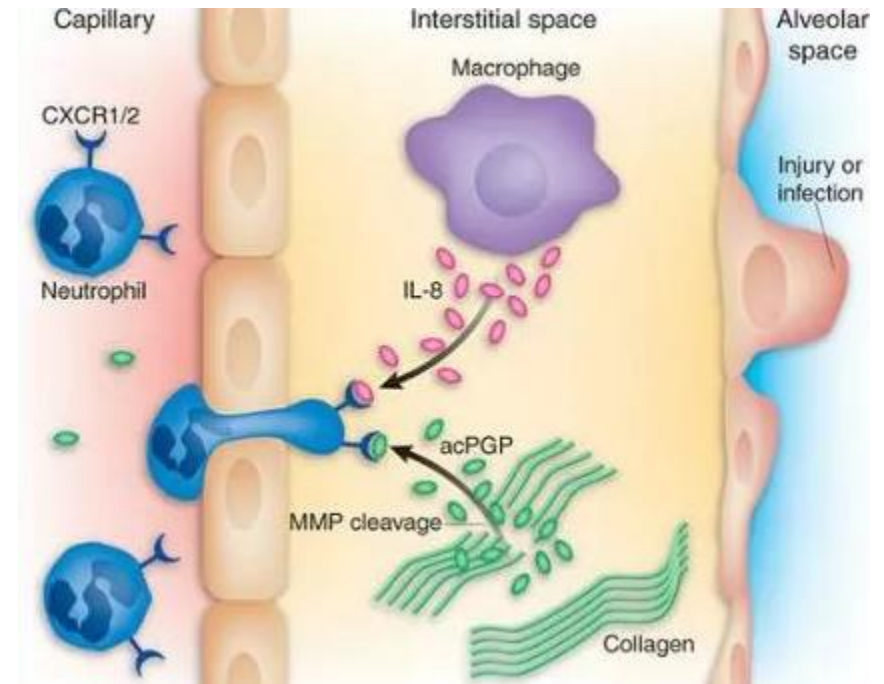
- ✓ interleukins,
- ✓ chemokines, and
- ✓ interferons.

Interleukins

- The interleukins were originally thought to be produced only by leukocytes (white blood cells) and to only stimulate leukocytes, thus the reasons for their name.
- Although interleukins are involved in modulating almost every function of the immune system, their role in the body is not restricted to immunity.
- Interleukins are also produced by and stimulate a variety of cells unrelated to immune defenses.

Chemokines

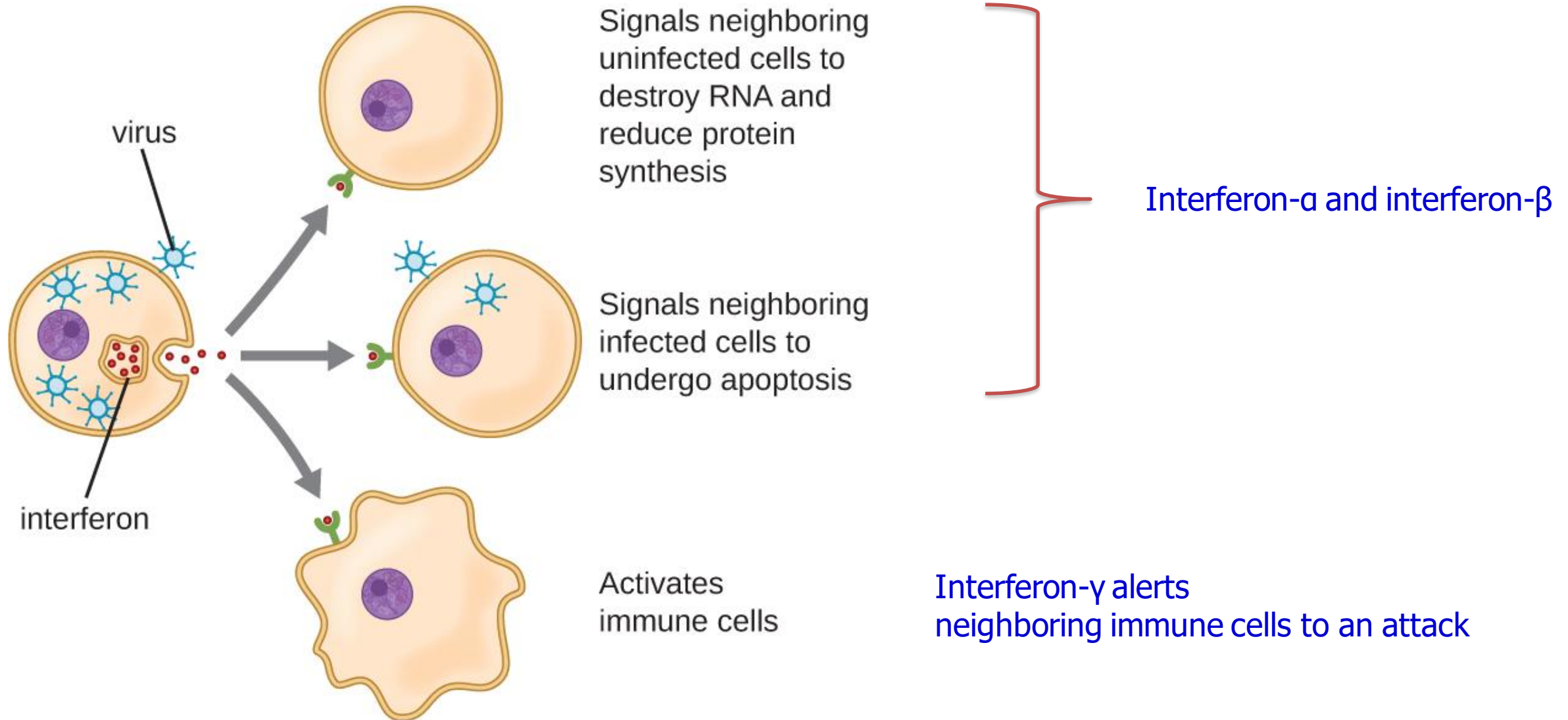
- ✓ A family of small proteins (8-10 kDa) secreted by cells to **control migration of nearby cells** (e.g. during tissue development or **immune response**).
- The chemokines are chemotactic factors that recruit leukocytes to sites of infection, tissue damage, and inflammation.
- In contrast to more general chemotactic factors, like complement factor C5a, chemokines are **very specific** in the subsets of leukocytes they recruit.
- ✓ playing a major role in the activation of host immune responses
- ✓ At least 47 different chemokines (4 groups): vary in specific function.



Interferons

- Interferons are a diverse group of immune signaling molecules and are especially important in **our defense against viruses**.
- **Type I interferons** (interferon- α and interferon- β) are produced and released by cells infected with the virus.
 - ✓ These interferons stimulate nearby cells to **stop the production of mRNA**, destroy RNA already produced, and reduce protein synthesis.
 - ✓ These cellular changes **inhibit viral replication** and production of mature virus, slowing the spread of the virus.
 - ✓ Also stimulate various immune cells involved in **viral clearance** to more aggressively attack virus-infected cells.
- Type II interferon (interferon- γ) is an important activator of immune cells.

Interferons



Contents

✓ Heme & Iron Metabolism

✓ Plasma proteins

✓ Blood buffer system

Buffering against pH Changes in Biological Systems

- Almost **every biological process is pH dependent**; a small change in pH produces a large change in the rate of the process.
- Cells and organisms maintain a specific and constant cytosolic pH, keeping biomolecules in their optimal ionic state, usually near pH 7.
- In multicellular organisms, the pH of extracellular fluids is also tightly regulated.
- **Constancy of pH** is achieved primarily by **biological buffers**: mixtures of weak acids and their conjugate bases.
- Biological buffering is illustrated by the **phosphate** and **carbonate buffering systems** of humans.

Metabolic Acids and Buffers

Metabolic acids and buffers

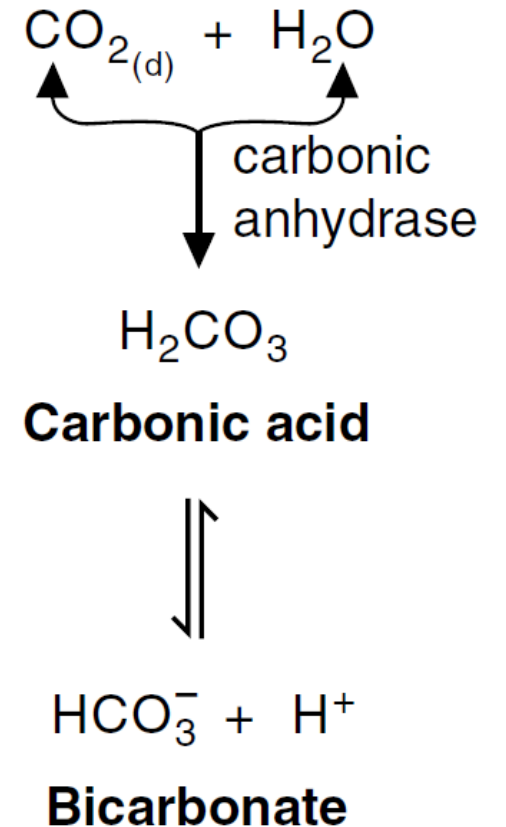
- The pH of the blood is normally maintained between 7.36 and 7.44, and intracellular pH at approximately 7.1 (between 6.9 and 7.4).
- The widest range of extracellular pH over which the metabolic functions of the liver, the beating of the heart, and the conduction of neural impulses can be maintained is 6.8 to 7.8.
- Thus, until the acid produced from metabolism can be excreted as CO₂ in **expired air** and as **ions in the urine**, it **needs to be buffered** in the body fluids.

Metabolic acids and buffers

- The major buffer systems in the body are:
 - ✓ the bicarbonate–carbonic acid buffer system, which operates principally in extracellular fluid;
 - ✓ the hemoglobin buffer system in red blood cells;
 - ✓ the phosphate buffer system in all types of cells; and
 - ✓ the protein buffer system of cells and plasma.

A. The Bicarbonate Buffer System

- The major source of metabolic acid in the body is the gas CO_2 , produced principally from **fuel oxidation** in the TCA cycle.
- Under normal metabolic conditions, the body generates more than 13 moles of CO_2 per day (approximately 0.51 kg).
- CO_2 dissolves in water and reacts with water to produce **carbonic acid**, H_2CO_3 , a reaction accelerated by the enzyme **carbonic anhydrase**.
- Carbonic acid is a weak acid that partially dissociates into **H^+** and bicarbonate anion, **HCO_3^-** .

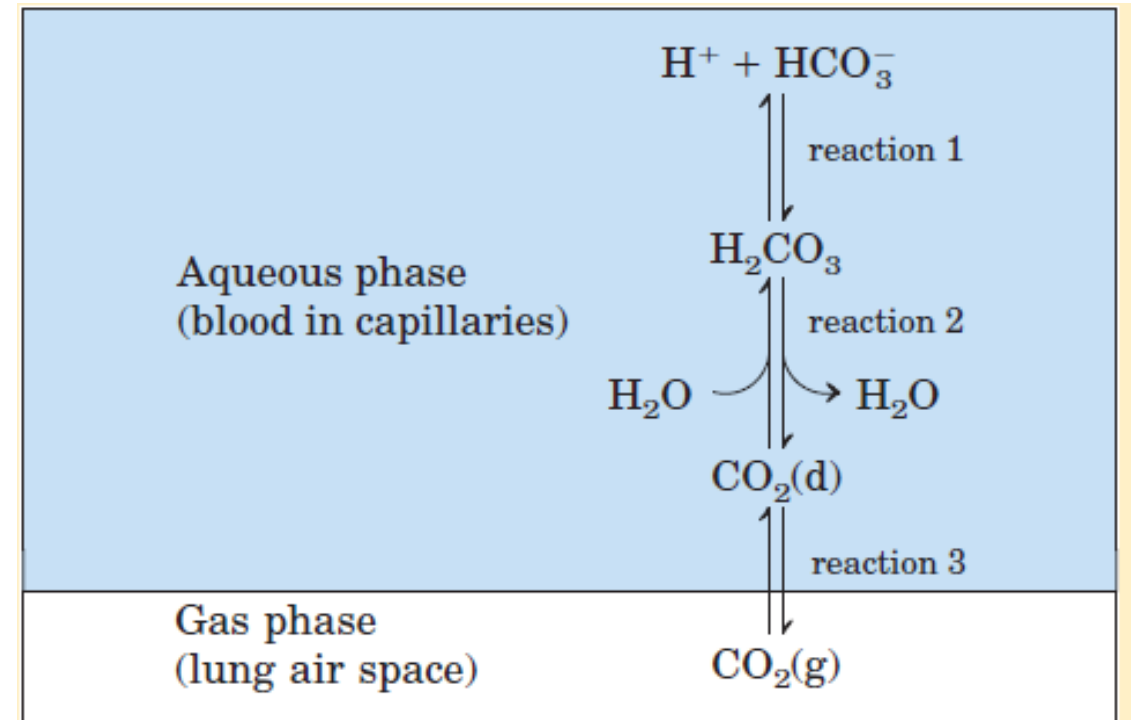


The Bicarbonate Buffer System

- Carbonic acid is both the major acid produced by the body, and its own buffer.
- The pKa of carbonic acid itself is only 3.8, so at the blood pH of 7.4 it is almost completely dissociated and theoretically unable to buffer and generate bicarbonate.
- However, carbonic acid can be replenished from CO_2 in body fluids and air because the concentration of dissolved CO_2 in body fluids is approximately 500 times greater than that of carbonic acid.
- The pKa for dissociation of bicarbonate anion (HCO_3^-) into H^+ and carbonate (CO_3^{2-}) is 9.8; therefore, **only trace amounts of carbonate exist in body fluids.**

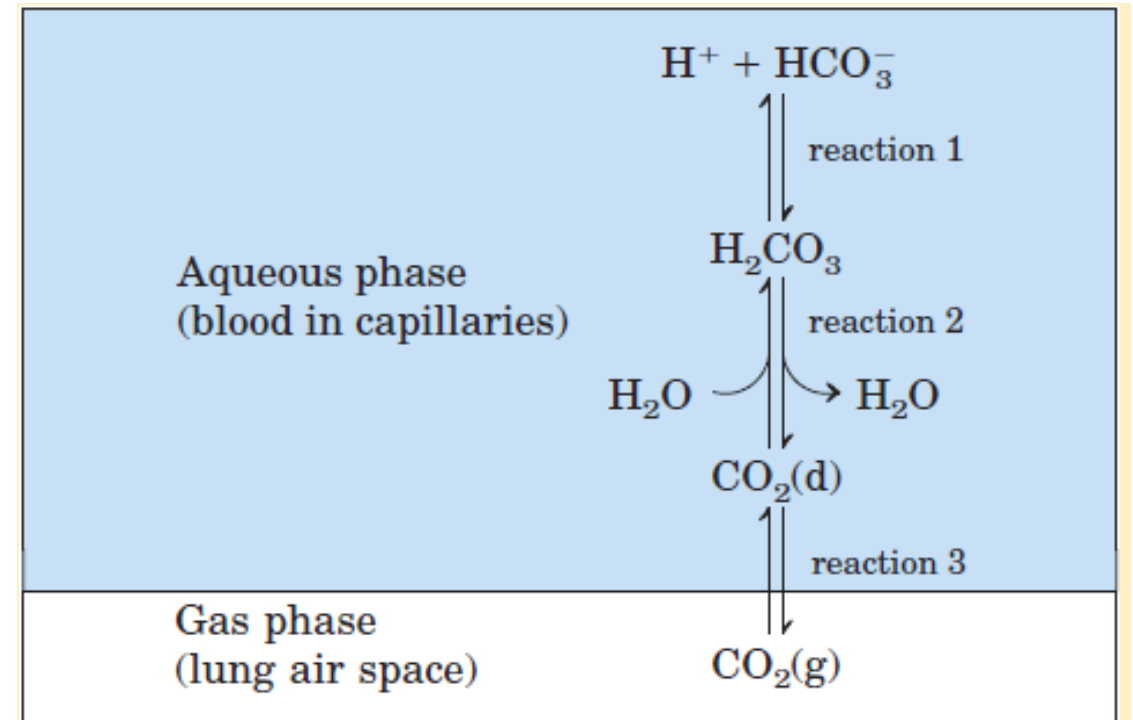
Blood, Lungs, and Buffer: The Bicarbonate Buffer System

- In animals with **lungs**, the bicarbonate buffer system is an effective physiological buffer near pH 7.4, because the H_2CO_3 of blood plasma is in equilibrium with a large reserve capacity of $\text{CO}_2(\text{g})$ in the air space of the lungs.
- This buffer system involves three reversible equilibria between gaseous CO_2 in the lungs and bicarbonate (HCO_3^-) in the blood plasma.

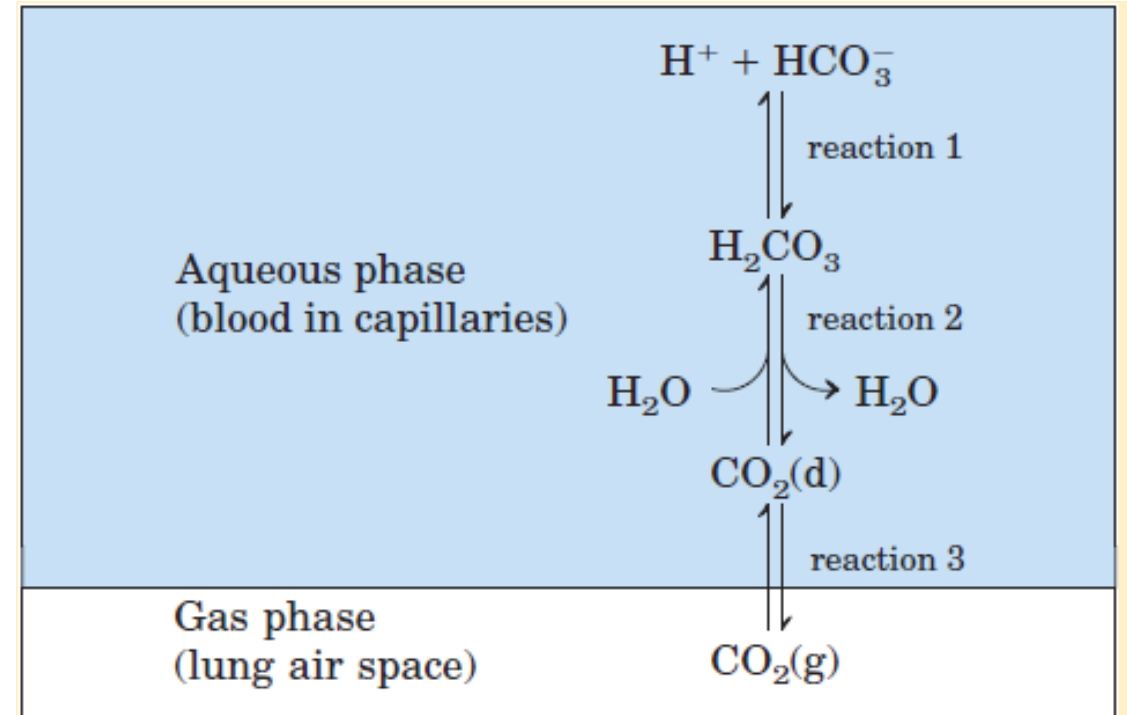


Blood, Lungs, and Buffer: The Bicarbonate Buffer System

- The CO_2 in the air space of the lungs is **in equilibrium** with the bicarbonate buffer in the blood plasma passing through the lung capillaries.
- Because the concentration of **dissolved CO_2** can be adjusted rapidly through changes in the rate of breathing, the bicarbonate buffer system of the blood is in near-equilibrium with a large potential reservoir of CO_2 .



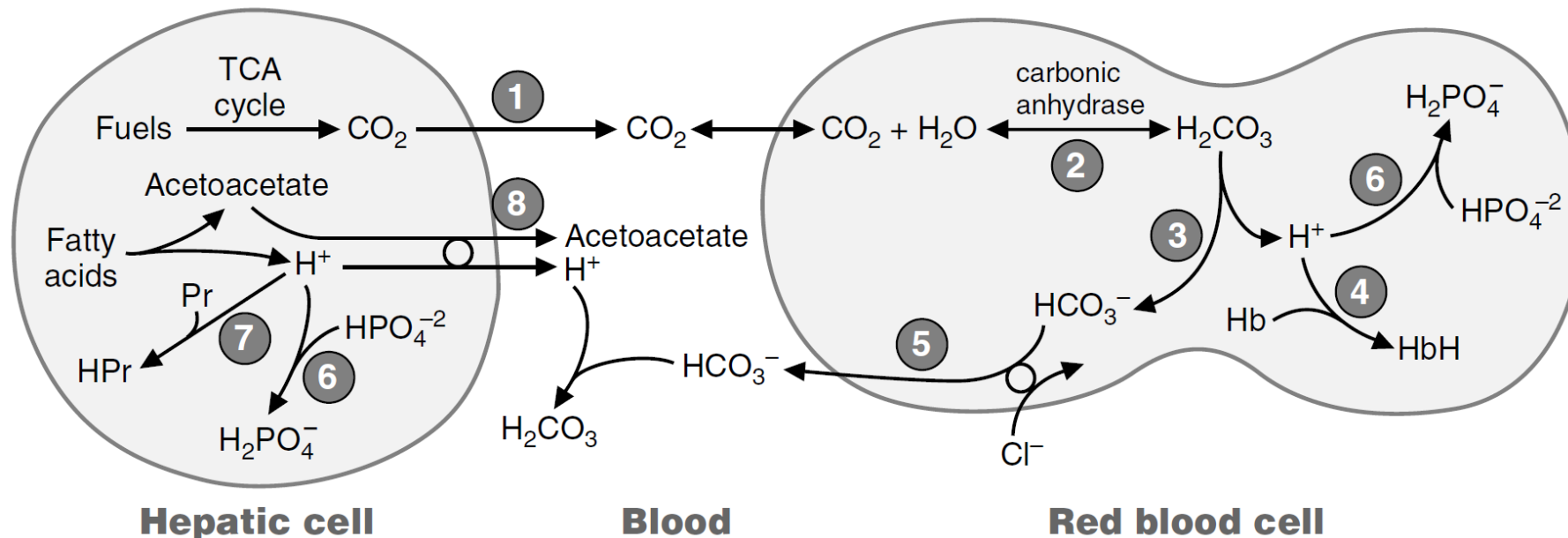
- Vigorous exercise:
 - ✓ **Lactic acid** increase
 - ✓ H^+ added to blood
 - ✓ $\text{CO}_2(\text{d})$ in blood plasma and $\text{CO}_2(\text{g})$
- Protein catabolism:
 - ✓ NH_3 increases
 - ✓ pH raised
 - ✓ More H_2CO_3



The rate of breathing—that is, the rate of inhaling and exhaling CO_2 —can quickly adjust these **equilibria to keep the blood pH nearly constant.**

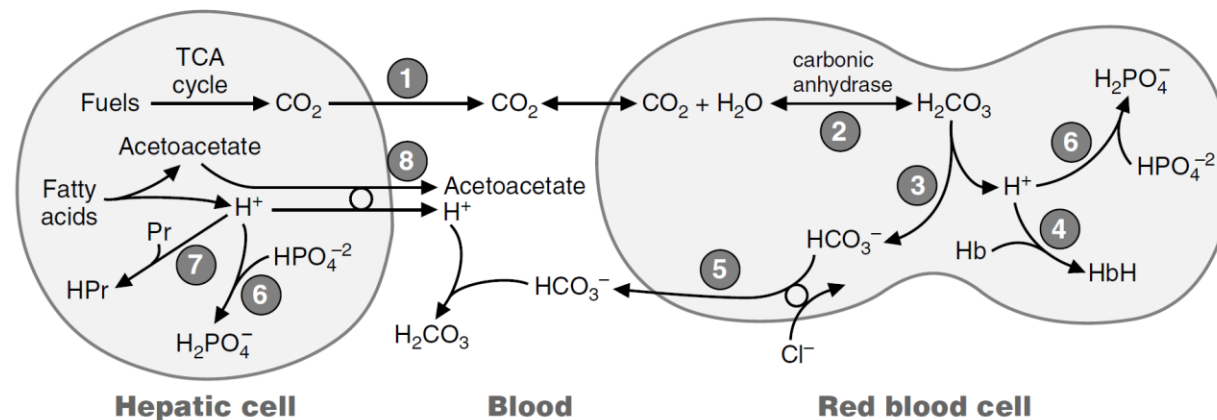
B. Bicarbonate and Hemoglobin in the Red Blood Cell

- The bicarbonate buffer system and hemoglobin in red blood cells cooperate in buffering the blood and transporting CO_2 to the lungs.
- Most of the CO_2 produced from tissue metabolism in the TCA cycle diffuses into the interstitial fluid and the blood plasma and then into red blood cells (**Fig circle 1**).



B. Bicarbonate and Hemoglobin in the Red Blood Cell

- Although no carbonic anhydrase can be found in blood plasma or interstitial fluid, the red blood cells contain high amounts of this enzyme, and CO_2 is rapidly converted to carbonic acid (H_2CO_3) within these cells (circle 2).
- As the carbonic acid dissociates (circle 3), the H^+ released is also buffered by combination with hemoglobin (Hb, circle 4). The side chain of the amino acid histidine in hemoglobin has a pK_a of 6.7 and is thus able to accept a proton.
- The bicarbonate anion is transported out of the red blood cell into the blood in exchange for chloride anion, and thus bicarbonate is relatively high in the plasma (circle 5).



. Distribution of ions in body fluids.

	ECF* mmol/L	ICF
Cations		
Na^+	145	12
K^+	4	150
Anions		
Cl^-	105	5
HCO_3^-	25	12
Inorganic Phosphate	2	100

*The content of inorganic ions is very similar in plasma and interstitial fluid, the two components of the extracellular fluid. (ECF, extracellular fluid; ICF, intracellular fluid).

B. Bicarbonate and Hemoglobin in the Red Blood Cell

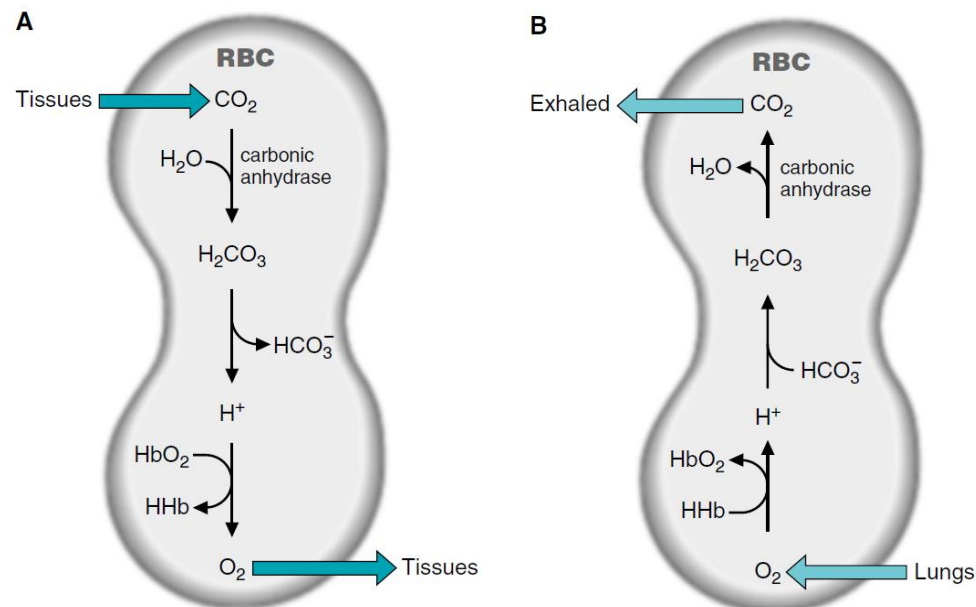
- As the red blood cell approaches the lungs, the direction of the equilibrium reverses. CO₂ is released from the red blood cell, causing more carbonic acid to dissociate into CO₂ and water and more hydrogen ions to combine with bicarbonate.
- Hemoglobin loses some of its hydrogen ions, a feature that allows it to bind oxygen more readily. Thus, the bicarbonate buffer system is intimately linked to the delivery of oxygen to tissues.
- The respiratory center within the hypothalamus, which controls the rate of breathing, is sensitive to changes in pH.
 - ✓ As the **pH falls**, individuals breathe more rapidly and expire more CO₂.
 - ✓ As the pH rises, they breathe more shallowly.
 - ✓ Thus, the rate of breathing contributes to the regulation of pH through its effects on the dissolved CO₂ content of the blood.

B. Bicarbonate and Hemoglobin in the Red Blood Cell

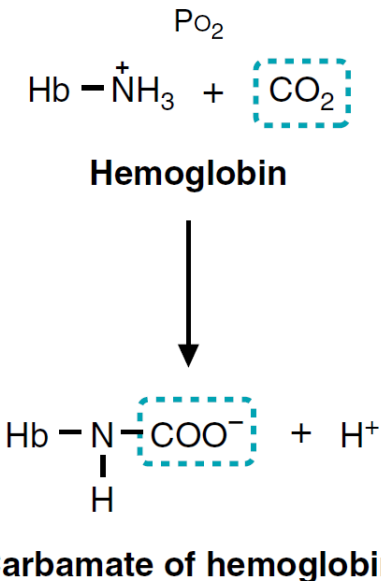
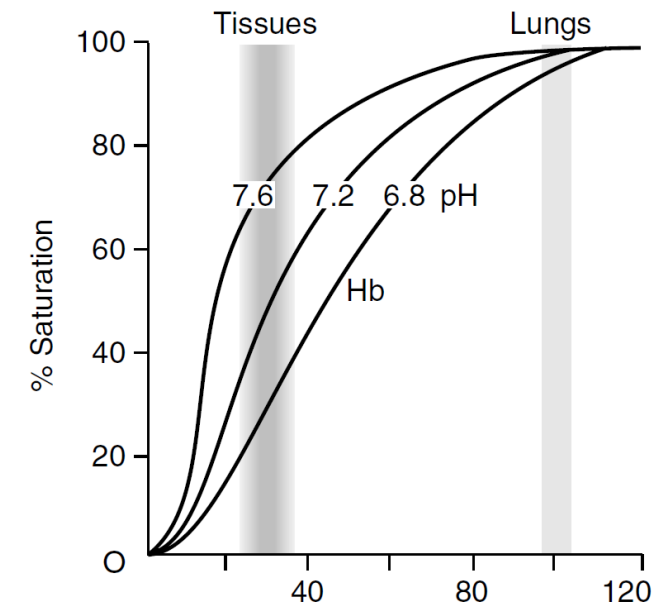
- Bicarbonate and carbonic acid, which diffuse through the capillary wall from the blood into the interstitial fluid, provide a major buffer for both plasma and interstitial fluid.
- However, blood differs from interstitial fluid in that the blood contains a high content of extracellular proteins, such as albumin, **which contribute to its buffering capacity** through amino acid side chains that are able to accept and release protons.
- The protein content of interstitial fluid is too low to serve as an effective buffer.

Agents that affect oxygen binding

- 2,3-Bisphosphoglycerate
- Proton Binding (Bohr effect)
- Carbon Dioxide



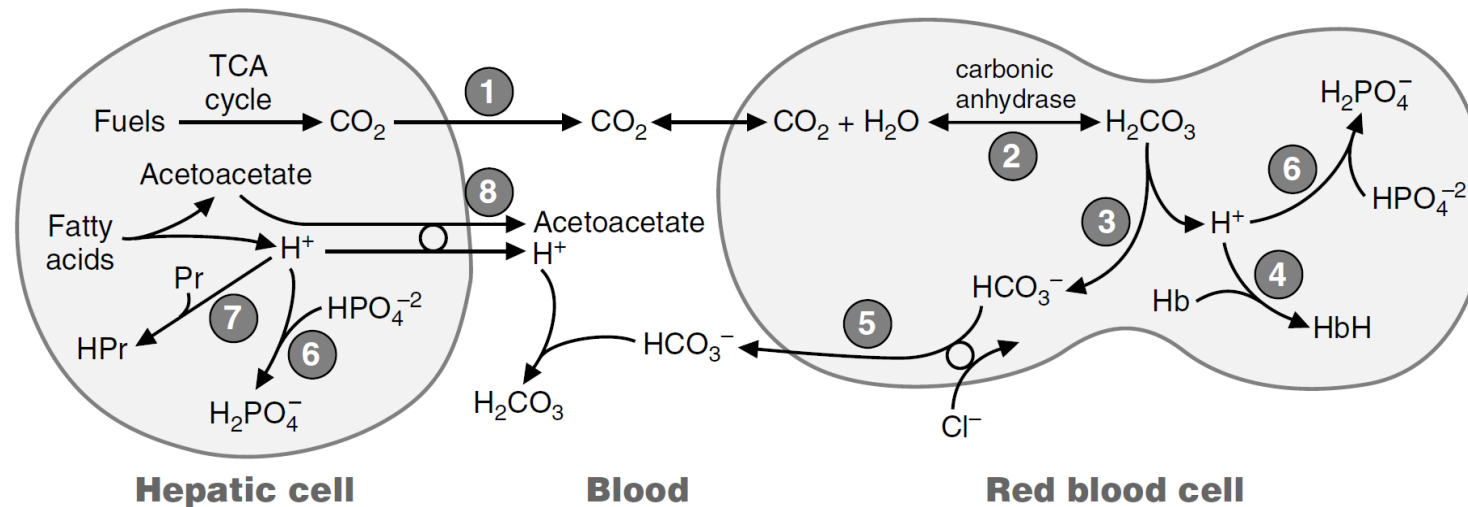
Solomon I Gizaw (PhD)



- In cases of severe protein malnutrition (kwashiorkor), the concentration of the plasma proteins decreases, as a result of which the **osmotic pressure** of the blood decreases.
- As a result, fluid is not drawn back to the blood and instead accumulates in the interstitial space (edema).
- The distended bellies of famine victims are the result of fluid accumulation in the extravascular tissues because of the severely decreased concentration of plasma proteins, particularly albumin.
- Albumin synthesis decreases fairly early under conditions of protein malnutrition.

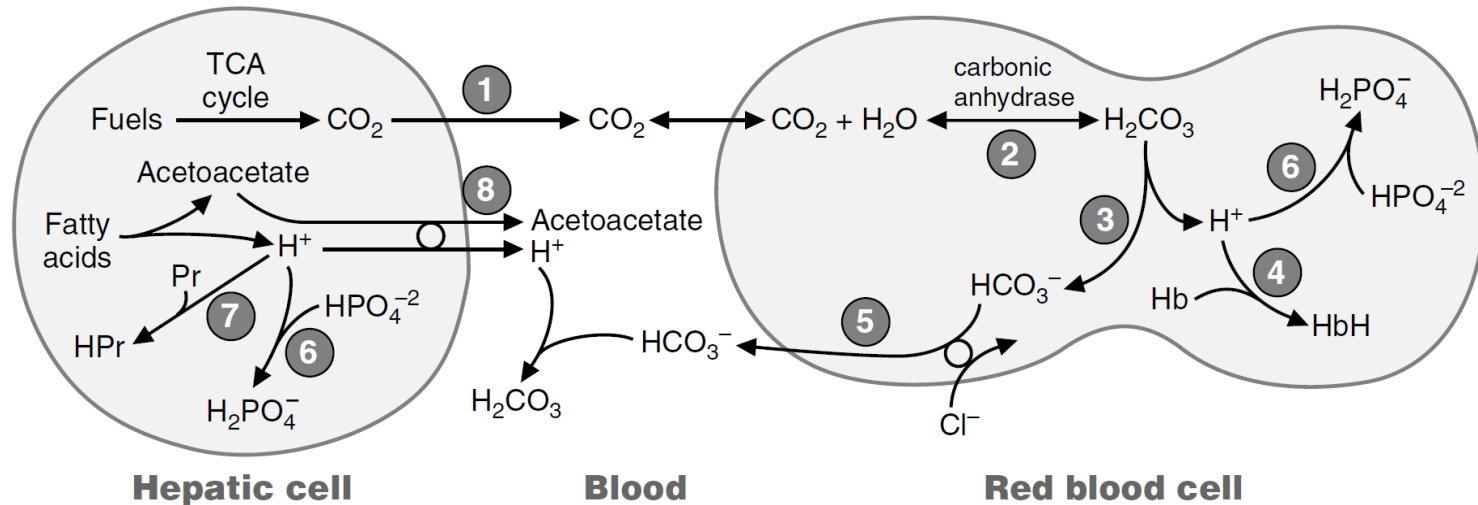
C. Intracellular pH

- Phosphate anions and proteins are the major buffers involved in maintaining a constant pH of intracellular fluids.
- The inorganic phosphate anion H_2PO_4^- dissociates to generate H^+ and the conjugate base, HPO_4^{2-} with a pK_a of 7.2 (**circle 6**).
- Thus, phosphate anions play a major role as an intracellular buffer in the red blood cell and in other types of cells, where their concentration is much higher than in blood and interstitial fluid (extracellular fluid)



C. Intracellular pH

- Organic phosphate anions, such as glucose 6-phosphate and ATP, also act as buffers.
- Intracellular fluid contains a high content of proteins that contain histidine and other amino acids that can accept protons, in a fashion similar to hemoglobin (**circle 7**).



Distribution of ions in body fluids.

	ECF* mmol/L	ICF
Cations		
Na ⁺	145	12
K ⁺	4	150
Anions		
Cl ⁻	105	5
HCO ₃ ⁻	25	12
Inorganic Phosphate	2	100

*The content of inorganic ions is very similar in plasma and interstitial fluid, the two components of the extracellular fluid. (ECF, extracellular fluid; ICF, intracellular fluid).

C. Intracellular pH

- The **transport of hydrogen ions out of the cell** is also important for the maintenance of a constant intracellular pH.
 - ✓ Metabolism produces a number of other acids in addition to CO_2 .
 - ✓ For example, the **metabolic acids acetoacetic acid** and **β -hydroxybutyric acid** are produced from fatty acid oxidation to ketone bodies in the liver, and
 - ✓ lactic acid is produced by **glycolysis** in muscle and other tissues.
- The pKa for most metabolic carboxylic acids is **below 5**, so these acids are completely dissociated at the pH of the blood and cellular fluid.
- **Metabolic anions are transported out of the cell together with H^+ (circle 8).**
- If the cell becomes too acidic, more H^+ is transported out in exchange for Na^+ ions by a different transporter.
- If the cell becomes too alkaline, more bicarbonate is transported out in exchange for Cl^- ions.

D. Urinary Hydrogen, Ammonium, and Phosphate Ions

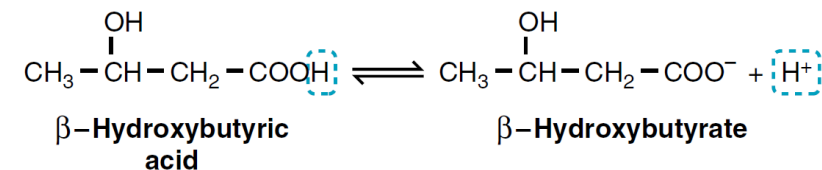
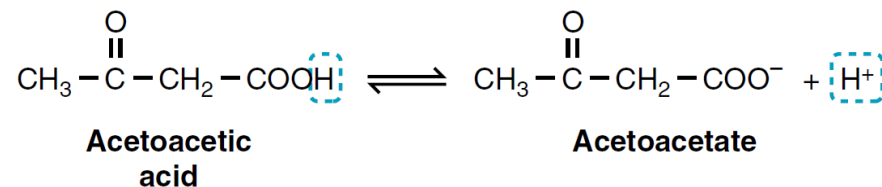
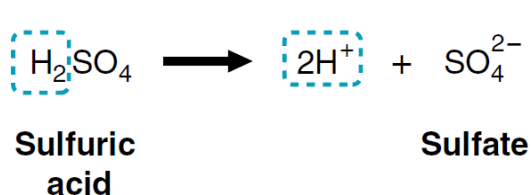
- The nonvolatile acid that is produced from body metabolism cannot be excreted as expired CO_2 and is **excreted in the urine**.
- Most of the nonvolatile acid hydrogen ion is excreted as undissociated acid that generally buffers the urinary pH between 5.5 and 7.0.
- A pH of 5.0 is the minimum urinary pH.**
- The acid secretion includes inorganic acids such as phosphate and ammonium ions, as well as uric acid, dicarboxylic acids, and tricarboxylic acids such as citric acid.

. Acids in the Blood of a Healthy Individual.

Acid	Anion	pK _a	Major Sources
Strong acid			
Sulfuric acid (H_2SO_4)	Sulfate SO_4^{-2}	Completely Dissociated	Dietary sulfate and S-containing amino acids
Weak acid			
Carbonic acid (R-COOH)	Bicarbonate (R-COO^-)	3.80	CO_2 from TCA cycle
Lactic acid (R-COOH)	Lactate (R-COO^-)	3.73	Anaerobic glycolysis
Pyruvic acid (R-COOH)	Pyruvate (R-COO^-)	2.39	Glycolysis
Citric acid (R-3COOH)	Citrate (R-3COO^-)	3.13; 4.76; 6.40	TCA cycle and diet (e.g., citrus fruits)
Acetoacetic acid (R-COOH)	Acetoacetate (R-COO^-)	3.62	Fatty acid oxidation to ketone bodies
β -Hydroxybutyric acid (R-COOH)	β -Hydroxybutyrate (R-COO^-)	4.41	Fatty acid oxidation to ketone bodies
Acetic acid (R-COOH)	Acetate (R-COO^-)	4.76	Ethanol metabolism
Dihydrogen phosphate (H_2PO_4^-)	Monohydrogen phosphate (HPO_4^{-2})	6.8	Dietary organic phosphates
Ammonium ion (NH_4^+)	Ammonia (NH_3)	9.25	Dietary nitrogen-containing compounds

D. Urinary Hydrogen, Ammonium, and Phosphate Ions

- One of the major sources of nonvolatile acid in the body is **sulfuric acid** (H_2SO_4).
 - ✓ Sulfuric acid is generated from the sulfate-containing compounds ingested in foods and from the **metabolism of the sulfur-containing amino acids**, cysteine and methionine.
 - ✓ It is a strong acid that is dissociated into H^+ and sulfate anion (SO_4^{2-}) in the blood and urine.
- Urinary excretion of H_2PO_4^- helps to remove acid.
- To maintain metabolic homeostasis, we must excrete the same amount of phosphate in the urine that we ingest with food as phosphate anions or organic phosphates such as **phospholipids**.
- Whether the phosphate is present in the urine as H_2PO_4^- or HPO_4^{2-} depends on the urinary pH and the pH of the blood.



D. Urinary Hydrogen, Ammonium, and Phosphate Ions

- Ammonium ions are major contributors to buffering urinary pH, but not blood pH.
- Ammonia (NH_3) is a base that combines with protons to produce ammonium (NH_4^+) ions ($\text{NH}_3 + \text{H}^+ \rightarrow \text{NH}_4^+$), a reaction that occurs with a pKa of 9.25.
- Ammonia is produced from **amino acid catabolism** or **absorbed through the intestine** and kept at very low concentrations in the blood because it is **toxic to neural tissues**.
- Cells in the kidney generate NH_4^+ and excrete it into the urine in proportion to the acidity (proton concentration) of the blood.
- As the renal tubular cells transport H^+ into the urine, they return bicarbonate anions to the blood.

E. Hydrochloric Acid

- Hydrochloric acid (HCl), also called gastric acid, is secreted by parietal cells of the stomach into the stomach lumen, where the **strong acidity denatures ingested proteins** so they can be degraded by digestive enzymes.
- When the stomach contents are released into the lumen of the small intestine, **gastric acid is neutralized by bicarbonate** secreted from pancreatic cells and by cells in the intestinal lining.

Contents COVERED

- ✓ Heme & Iron Metabolism
- ✓ Plasma proteins
- ✓ Blood buffer system

Please do not limit your reading
to this note.

REFER TO BOOKS!

Thank you for your kind attention